Chapter 22

Gauss's Law

Chapter Objectives

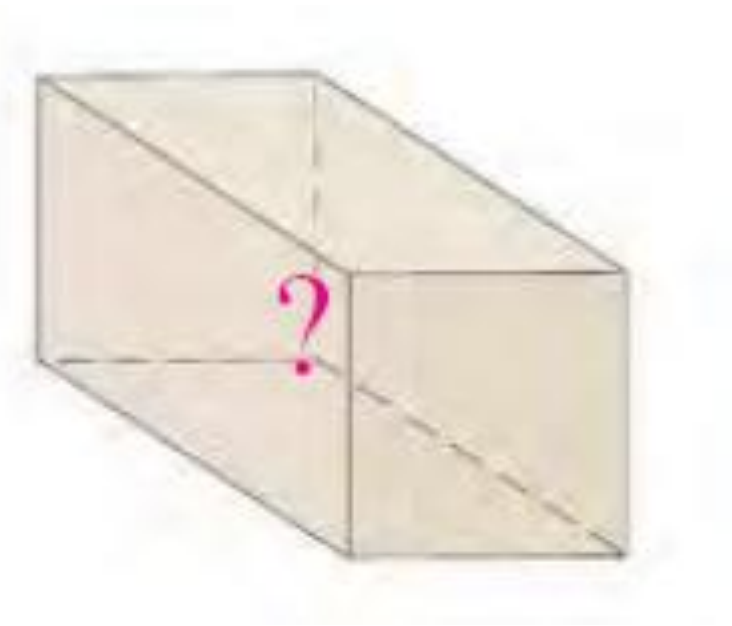
- To know that Gauss's law uses symmetric considerations to simplify electric-field calculations.
- To understand that Gauss's law is a relation between the field at all the points on an imaginary surface enclosing any general distribution of charge and the total charge enclosed within the surface.
- To gain deeper insights into electric charge distribution and electric fields.

Chapter Outline

1. Charge and Electric Flux
2. Calculating Electric Flux
3. Gauss's Law
4. Applications of Gauss's Law
5. Charges on Conductors

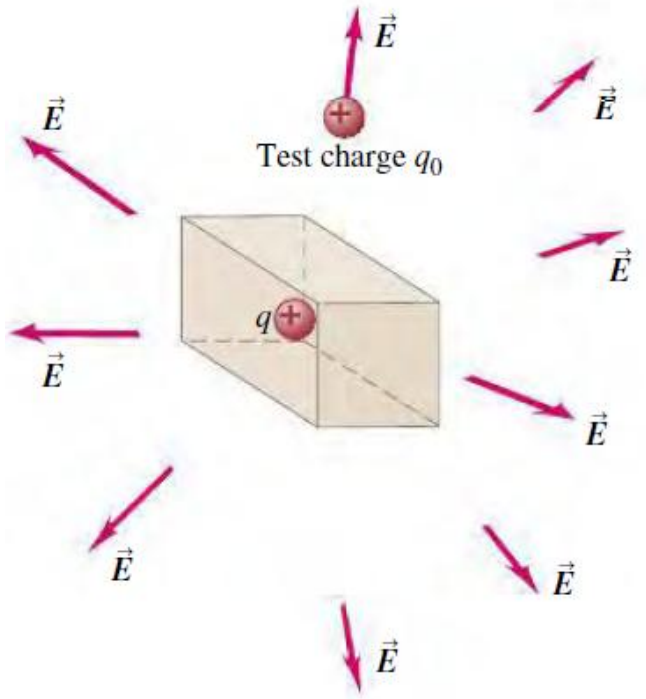
22.1 Charge and Electric Flux

- The figure shows a closed surface (a box), containing an unknown amount of charge.



22.1 Charge and Electric Flux

- Using a test charge outside the box to probe the amount of charge inside the box.



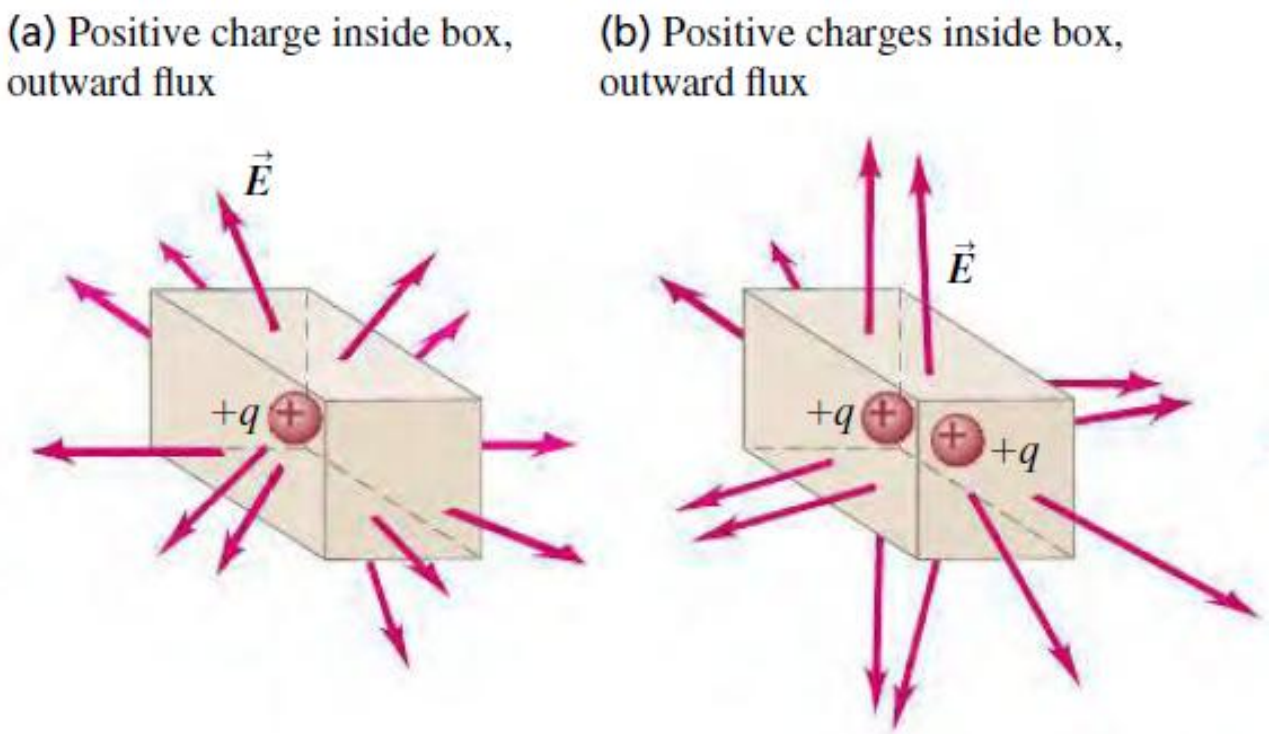
22.1 Charge and Electric Flux

- By measuring the force $\vec{F}$ experienced by the test charge at different positions, the electric field $\vec{E} = \frac{\vec{F}}{q_0}$ outside the box can be mapped.

- From this $3-D$ map, the exact value of the point charge inside the box can be found.

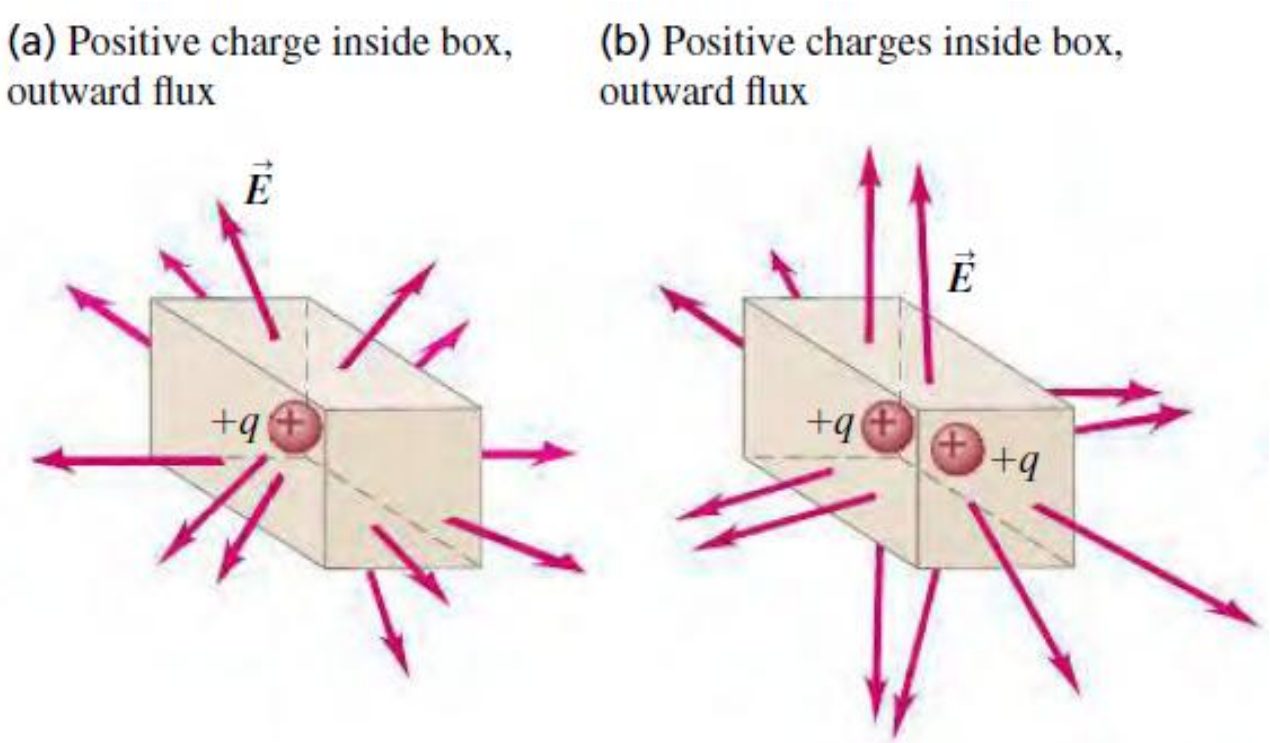
22.1 Charge and Electric Flux

- Figure shows the electric field vectors (electric flux) caused by positive charges.



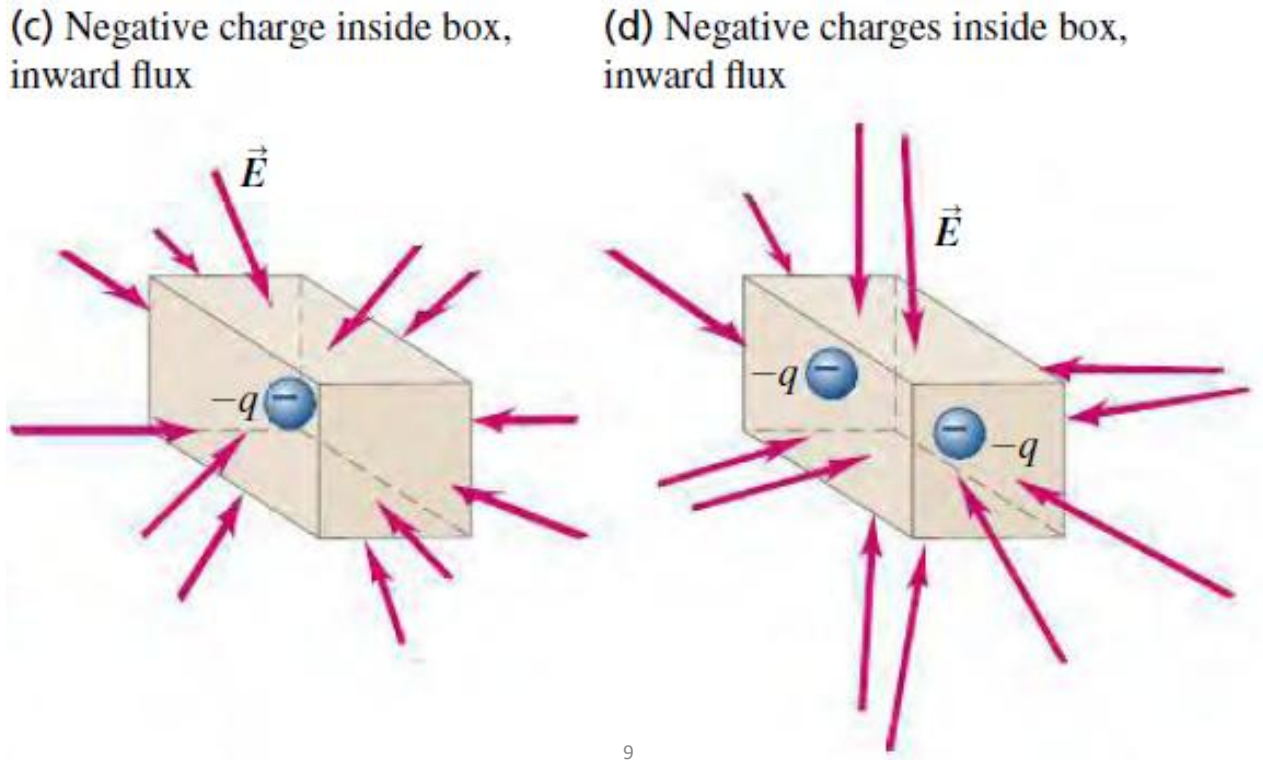
22.1 Charge and Electric Flux

- Positive electric point charge: outward electric flux



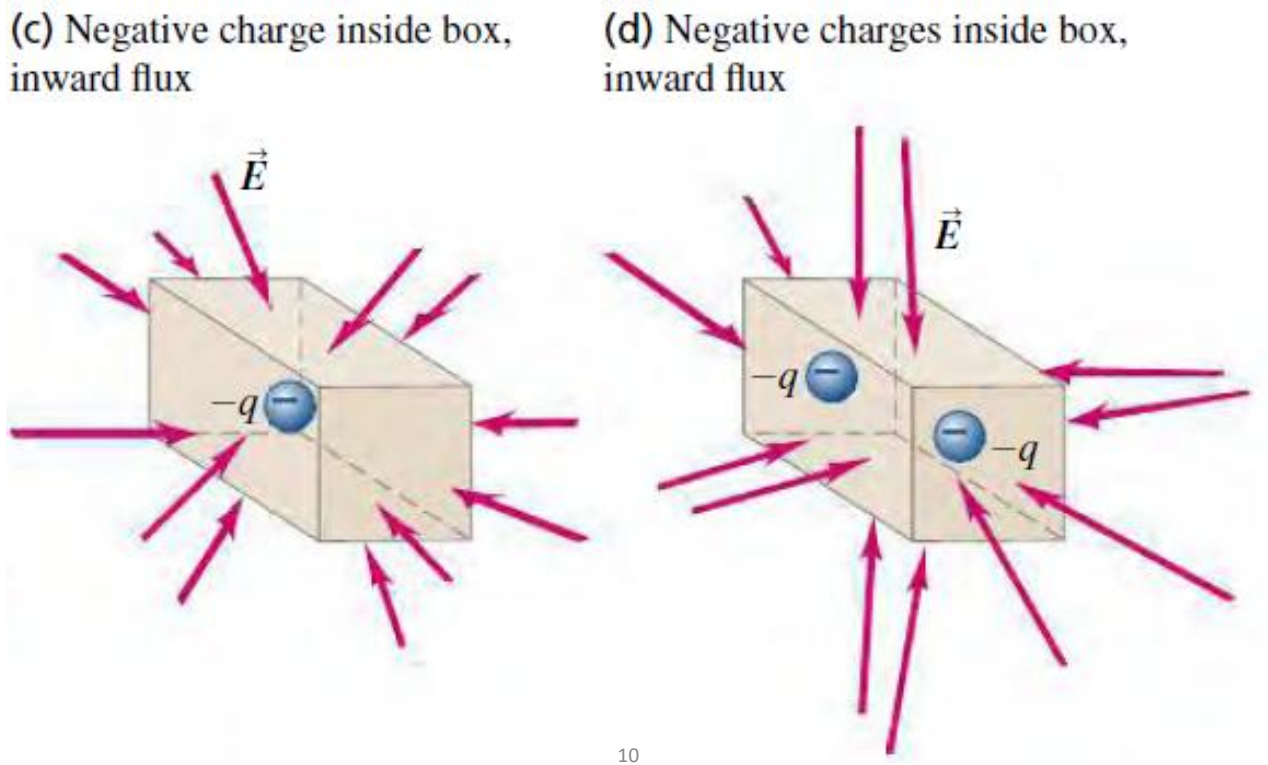
22.1 Charge and Electric Flux

- Figure shows the electric field vectors (electric flux) caused by negative charges.



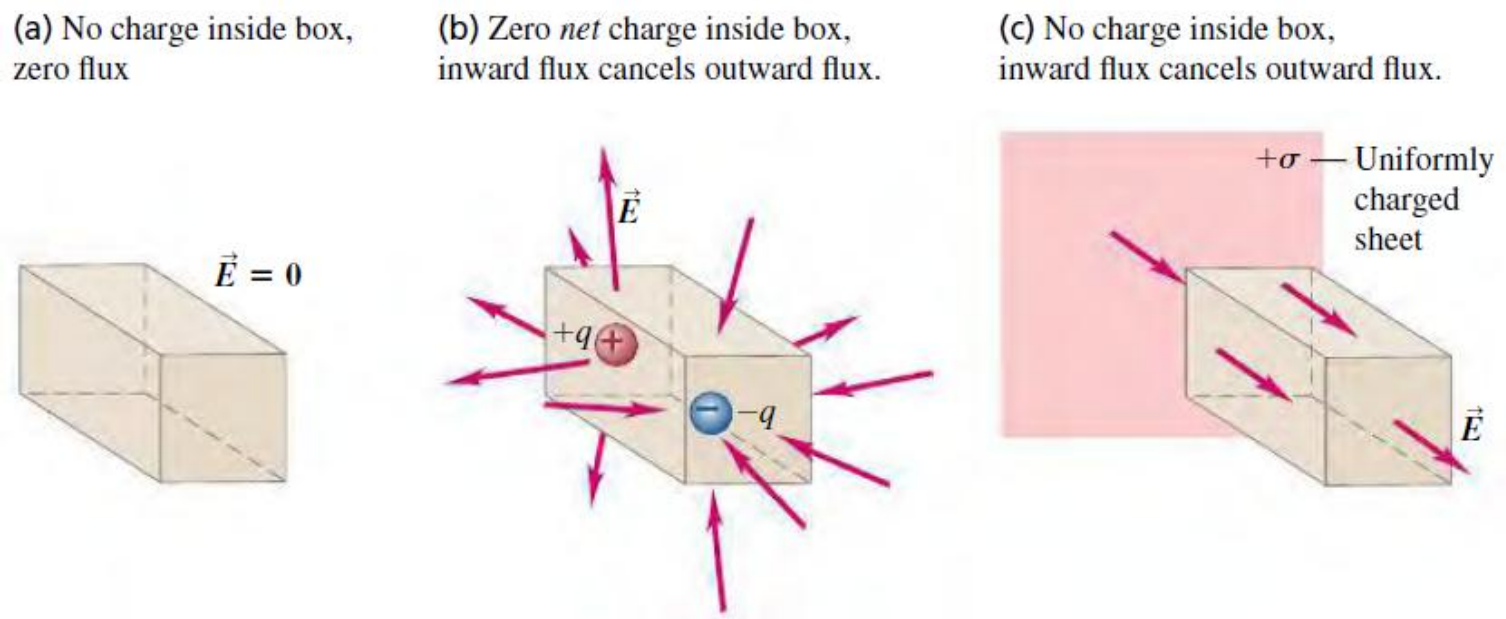
22.1 Charge and Electric Flux

- Negative electric point charge: inward electric flux



22.1 Charge and Electric Flux

- When there is no net charge inside the box, there is no net electric flux into or out of the box.
- In the figure below, three such cases are shown.



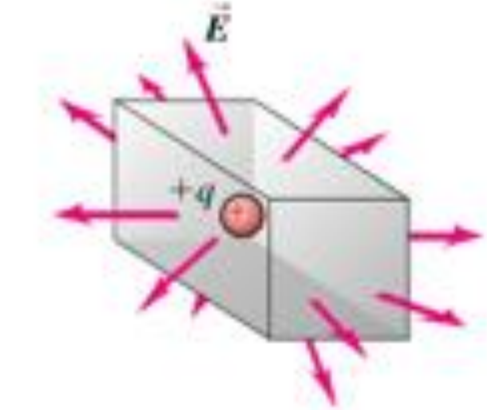
22.1 Charge and Electric Flux

- Electric flux is defined as follows: For each face of the box, take the product of the average perpendicular component of $\vec{E}$ and the area of that face. Then add up the results from all faces of the box.

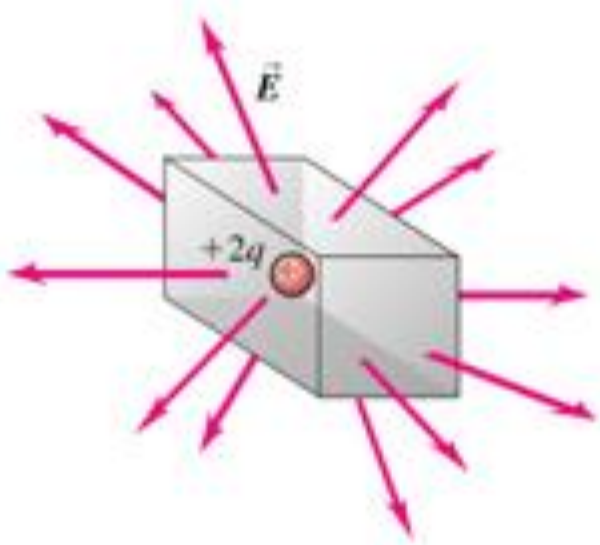
- Thus, the net electric flux through the surface of the box is directly proportional to the magnitude of the net charge enclosed by it, independent of the size of the box.

22.1 Charge and Electric Flux

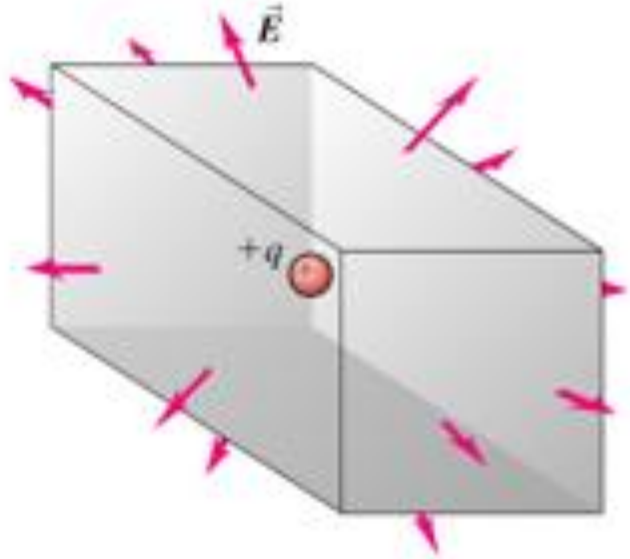
- The figure below illustrates this principle.



(a) Positive charge inside box, outward flux



(b) Same box as (a), twice the positive charge: twice the outward flux of (a)



(c) Box twice the dimensions of (a), same positive charge: same outward flux as (a)

Qualitative statement of Gauss's law

1. Whether there is a net outward or inward electric flux through a closed surface depends on the sign of the enclosed charge.

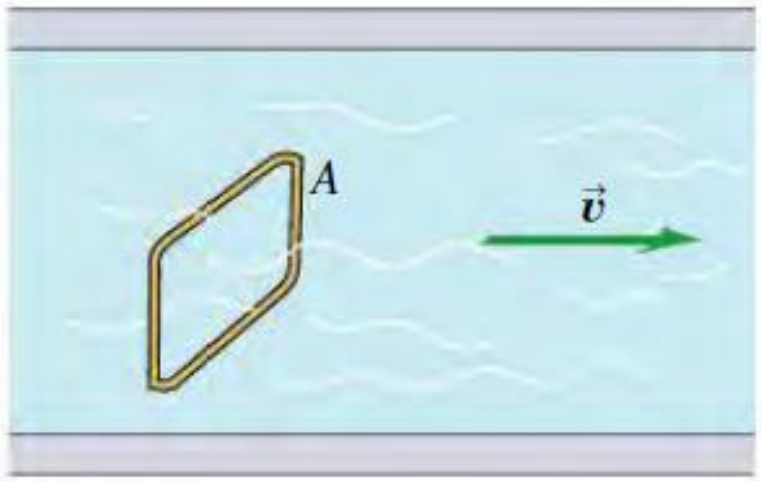
2. Charges outside the surface do not give a net electric flux through the surface.

3. The net electric flux is directly proportional to the net amount of charge enclosed within the surface but is otherwise independent of the size of the closed surface.

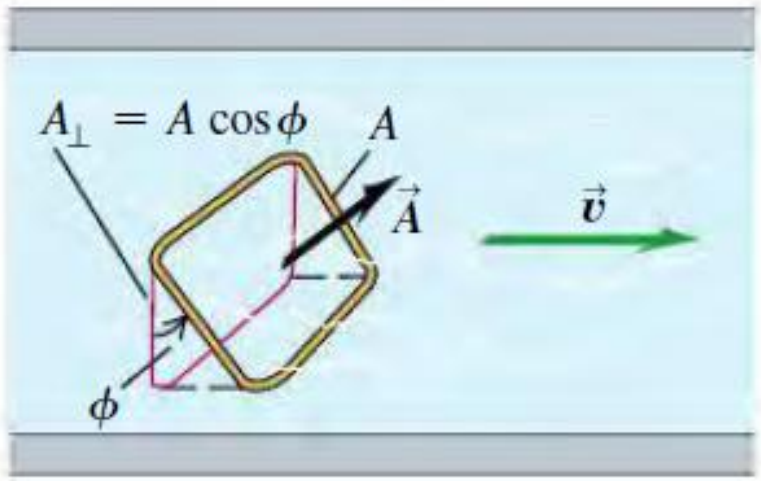
22.2 Calculating Electric flux

The volume flow rate of fluid through the wire rectangle
(a) is vA when the area of the rectangle is perpendicular to $\vec{v}$ and
(b) is $vA\cos\phi$ when the rectangle is tilted at an angle ϕ.

(a) A wire rectangle in a fluid



(b) The wire rectangle tilted by an angle ϕ

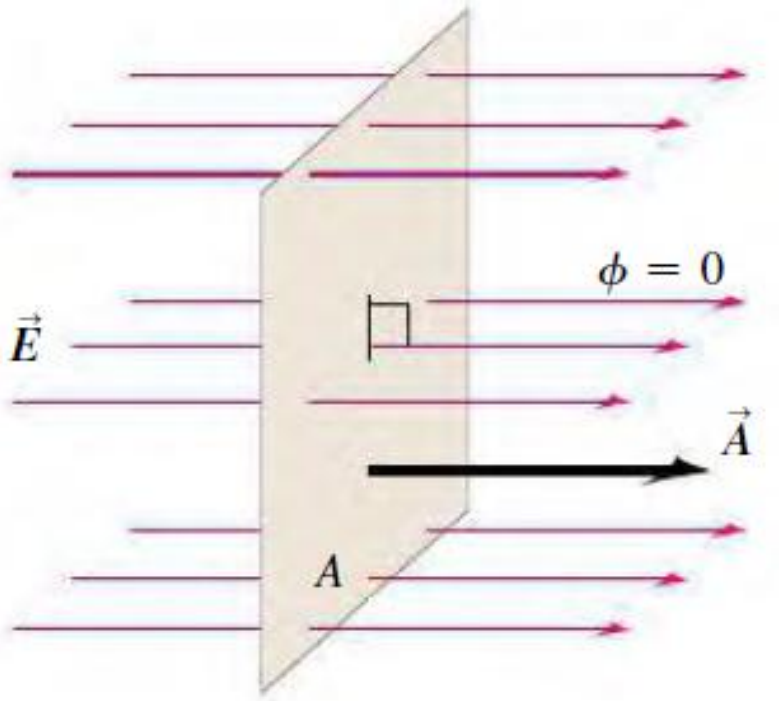


22.2 Calculating Electric flux

- Using the analogy between electric field and fluid flow, we replace the fluid velocity $\vec{v}$ by the electric field $\vec{E}$ in our electric flux calculations.

- The figure in the next slide shows how to calculate the electric flux.

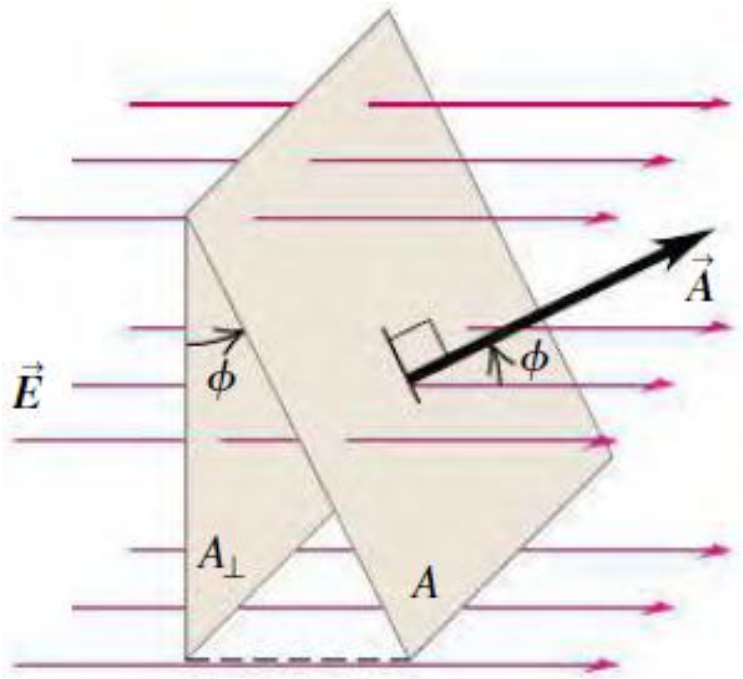
22.2 Calculating Electric flux



a) Surface is face-on to electric field:

- $\vec{E}$ and $\vec{A}$ are parallel (the angle between $\vec{E}$ and $\vec{A}$ is $\phi = 0$).
- The flux $\Phi_E = \vec{E} \cdot \vec{A} = EA$

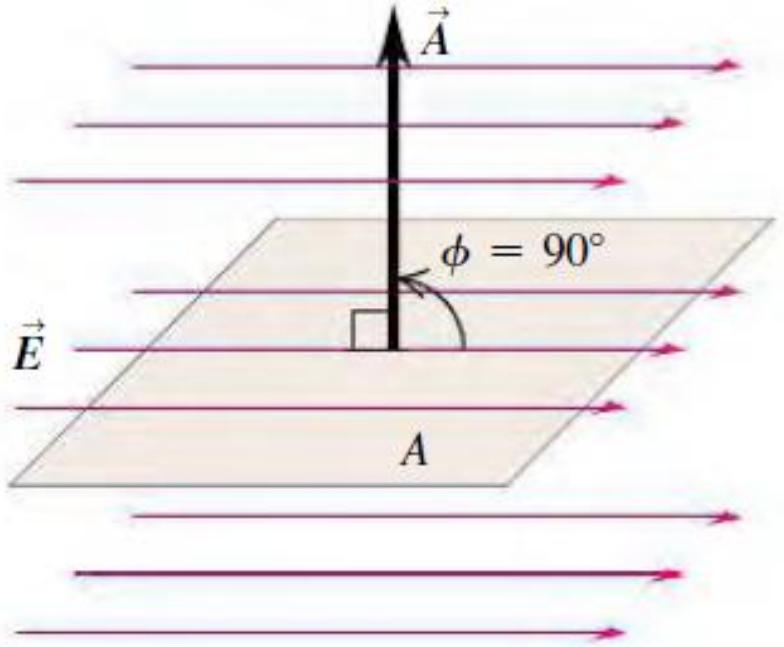
22.2 Calculating Electric flux



b) Surface is tilted from a face-on orientation by an angle ϕ:

- the angle between $\vec{E}$ and $\vec{A}$ is ϕ.
- The flux $\Phi_E = \vec{E} \cdot \vec{A} = EA\cos\phi$

22.2 Calculating Electric flux



c) Surface is edge-on to electric field:

- $\vec{E}$ and $\vec{A}$ are perpendicular (the angle between $\vec{E}$ and $\vec{A}$ is $\phi = 90°$).
- The flux $\Phi_E = \vec{E} \cdot \vec{A} = EA\cos 90° = 0$.

22.2 Calculating Electric flux

- For a uniform electric field,

$$\Phi_E = EA\cos\phi$$
$$= E_\perp A$$
$$= \vec{\boldsymbol{E}} \cdot \vec{\boldsymbol{A}}$$

(electric flux for a uniform field $\vec{\boldsymbol{E}}$, flat surface)

- The SI unit for electric flux is $1\ N \cdot m^2/C$.

22.2 Calculating Electric flux

- Vector area $\vec{A}$: $\vec{A} = A\hat{n}$

  The direction of $\hat{n}$ is defined as outward, with the corresponding flux moving out of a closed surface.

- Outward electric flux corresponds to a positive value of Φ_E.

- Inward electric flux corresponds to a negative value of Φ_E.

22.2 Calculating Electric flux

General definition of electric flux:

$$\Phi_E = \int E\cos\phi\, dA = \int E_\perp\, dA = \int \vec{\boldsymbol{E}}\cdot d\vec{\boldsymbol{A}} \qquad (22.5)$$

(general definition of electric flux)

Example 22.1 Electric flux through a disk

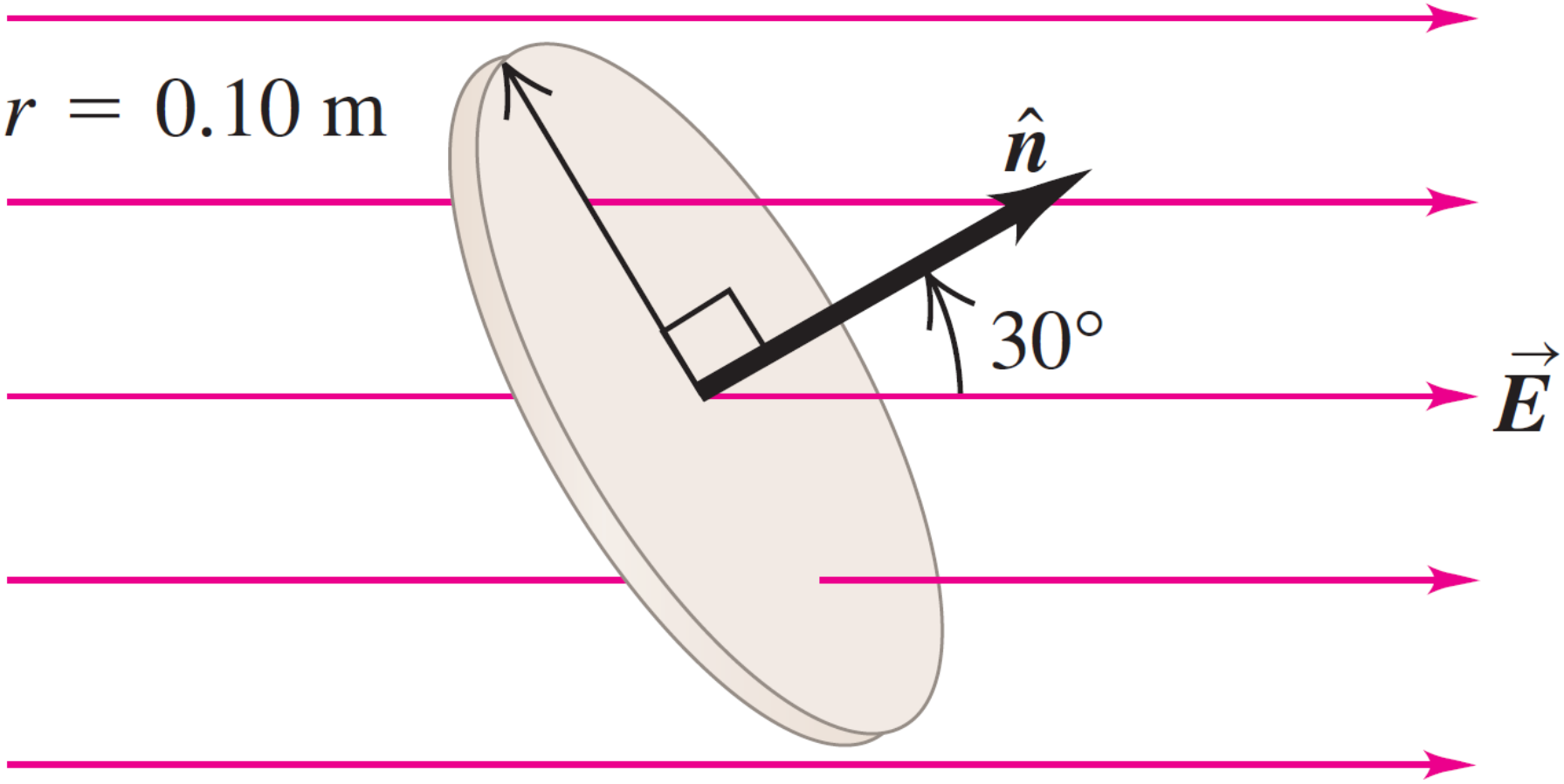
A disk with radius $0.10\ m$ is oriented with its normal unit vector $\hat{\boldsymbol{n}}$ at 30° to a uniform electric field $\vec{\boldsymbol{E}}$ of magnitude $2.0 \times 10^3\ N/C$. (Since this disk is not a closed surface, it has no "inside" or "outside". That's why we have to specify the direction of $\hat{\boldsymbol{n}}$ in the figure.)

a) What is the electric flux through the disk?

b) What is the flux through the disk if it is turned so that its normal is perpendicular to $\vec{\boldsymbol{E}}$?

c) What is the flux through the disk if $\hat{\boldsymbol{n}}$ is parallel to $\vec{\boldsymbol{E}}$?

Example 22.1 Electric flux through a disk



Identify and Set Up

The problem is about a flat surface in a uniform electric field. We can calculate the electric flux using $\Phi_E = EA\cos\phi$.

The electric flux Φ_E through a disk depends on the angle between its normal $\hat{\boldsymbol{n}}$ and the electric field $\vec{\boldsymbol{E}}$.

Execute

a) The area is $A = \pi(0.10\ m)^2 = 0.0314\ m^2$ and the angle between $\vec{\boldsymbol{E}}$ and $\vec{\boldsymbol{A}} = A\hat{\boldsymbol{n}}$ is $\phi = 30°$, so

$$\begin{aligned}\Phi_E &= EA\cos\phi \\ &= \left(2.0 \times 10^3\ N/C\right)\left(0.0314\ m^2\right)(\cos 30°) \\ &= 54\ N \cdot m^2/C\end{aligned}$$

b) The normal to the disk is now perpendicular to $\vec{\boldsymbol{E}}$, so $\phi = 90°$, $\cos\phi = 0$, and $\Phi_E = 0$. There is no flux through the disk.

Execute

c) The normal to the disk is parallel to $\vec{E}$, so $\phi = 0$, $\cos\phi = 1$:

$$\begin{aligned}\Phi_E &= EA\cos\phi \\ &= \left(2.0\times10^3\ N/C\right)\left(0.0314\ m^2\right)(1) \\ &= 63\ N\cdot m^2/C\end{aligned}$$

Evaluate

Our answer to part (b) is smaller than that to part (a), which is in turn smaller than that to part (c). Is all this as it should be?

Example 22.2 Electric flux through a cube

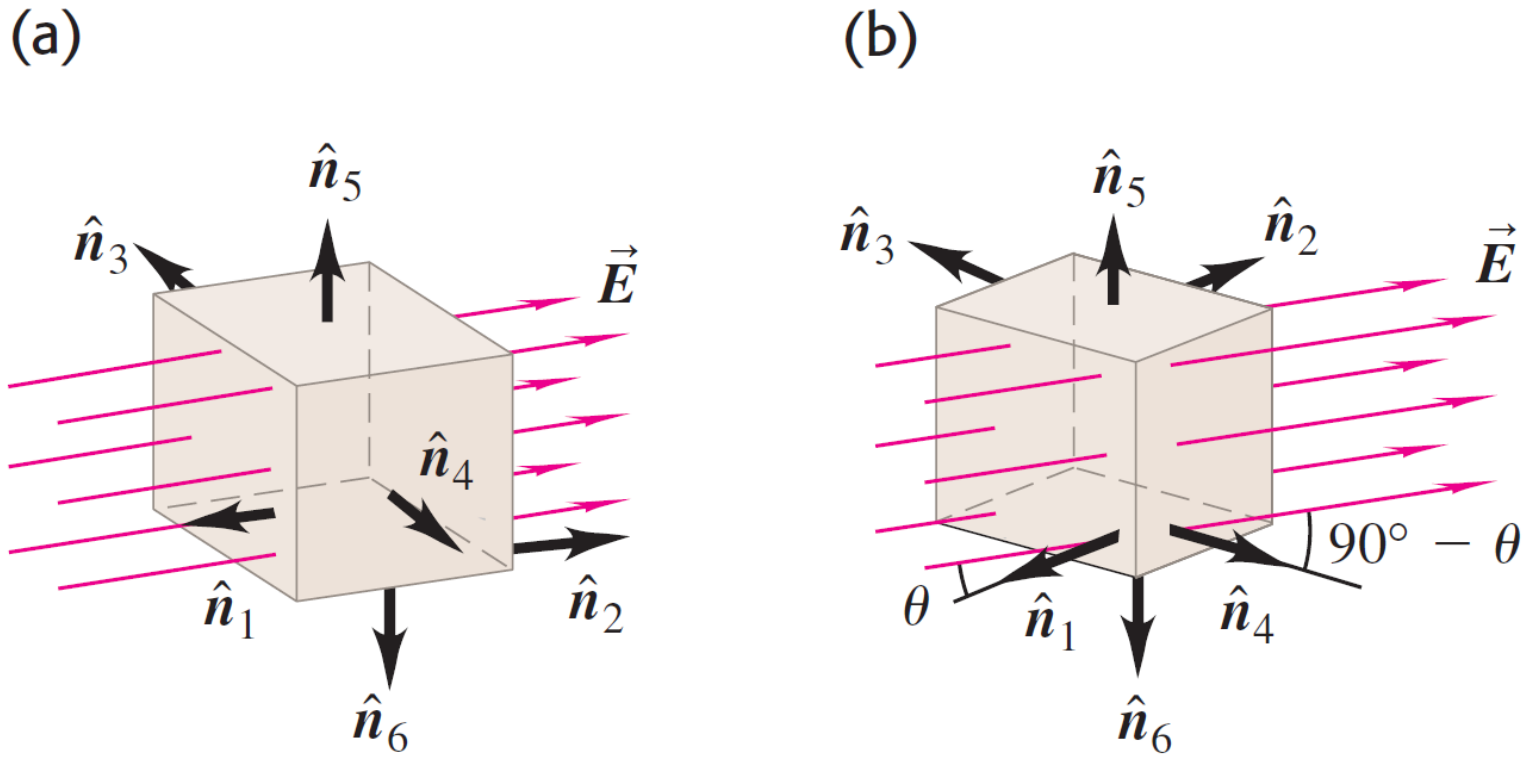
A cube of side L is placed in a region of uniform electric field $\vec{E}$.

Find the electric flux through each face of the cube and the total flux through the cube when

a) it is oriented with two of its faces perpendicular to the field $\vec{E}$, as in the Fig. a in the next slide;

b) when the cube is turned by an angle θ, as in the Fig. b in the next slide.

Example 22.2 Electric flux through a cube



(Electric flux of a uniform field $\vec{E}$ through a cubical box of side L in two orientations)

Identify and Set Up

Since $\vec{E}$ is uniform and each of the six faces of the cube is a flat surface, we find the flux through each face.

We then calculate the total flux through the cube by adding the six individual fluxes.

Execute

The unit vectors $\hat{\boldsymbol{n}}_1$ through $\hat{\boldsymbol{n}}_6$ for each face are shown in the figure; the direction of each unit vector is outward from the closed surface of the cube.

The angle between $\vec{\boldsymbol{E}}$ and $\hat{\boldsymbol{n}}_1$ is $180°$; the angle between $\vec{\boldsymbol{E}}$ and $\hat{\boldsymbol{n}}_2$ is $0°$.

Each face of the cube has area L^2, so the fluxes through each of the faces are

Execute

$$\Phi_{E_1} = \vec{\boldsymbol{E}} \cdot \hat{\boldsymbol{n}}_1 A = EL^2 \cos 180° = -EL^2$$

$$\Phi_{E_2} = \vec{\boldsymbol{E}} \cdot \hat{\boldsymbol{n}}_2 A = EL^2 \cos 0° = +EL^2$$

$$\Phi_{E_3} = \Phi_{E_4} = \Phi_{E_5} = \Phi_{E_6} = EL^2 \cos 90° = 0$$

Execute

a) The flux is negative on face 1, where $\vec{E}$ is directed into the cube, and positive on face 2, where $\vec{E}$ is directed out of the cube.

The total flux through the cube is the sum of the fluxes through the six faces:

$$\begin{aligned}\Phi_E &= \Phi_{E_1} + \Phi_{E_2} + \Phi_{E_3} + \Phi_{E_4} + \Phi_{E_5} + \Phi_{E_6} \\ &= -EL^2 + EL^2 + 0 + 0 + 0 + 0 \\ &= 0\end{aligned}$$

Execute

b) Those fluxes through faces 1 and 3 are negative, since $\vec{\boldsymbol{E}}$ is directed into those faces; the field is directed out faces 2 and 4, so the fluxes through those faces are positive.

We find:

$$\Phi_{E_1} = \vec{\boldsymbol{E}} \cdot \hat{\boldsymbol{n}}_1 A = EL^2 \cos(180° - \theta) = -EL^2 \cos\theta$$

$$\Phi_{E_2} = \vec{\boldsymbol{E}} \cdot \hat{\boldsymbol{n}}_2 A = +EL^2 \cos\theta$$

Execute

$$\Phi_{E_3} = \vec{\boldsymbol{E}} \cdot \hat{\boldsymbol{n}}_3 A = EL^2 \cos(90° + \theta) = -EL^2 \sin\theta$$

$$\Phi_{E_4} = \vec{\boldsymbol{E}} \cdot \hat{\boldsymbol{n}}_4 A = EL^2 \cos(90° - \theta) = +EL^2 \sin\theta$$

$$\Phi_{E_5} = \Phi_{E_6} = EL^2 \cos 90° = 0$$

The total flux through the surface of the cube :

$$\begin{aligned}\Phi_E &= \Phi_{E_1} + \Phi_{E_2} + \Phi_{E_3} + \Phi_{E_4} + \Phi_{E_5} + \Phi_{E_6} \\ &= 0\end{aligned}$$

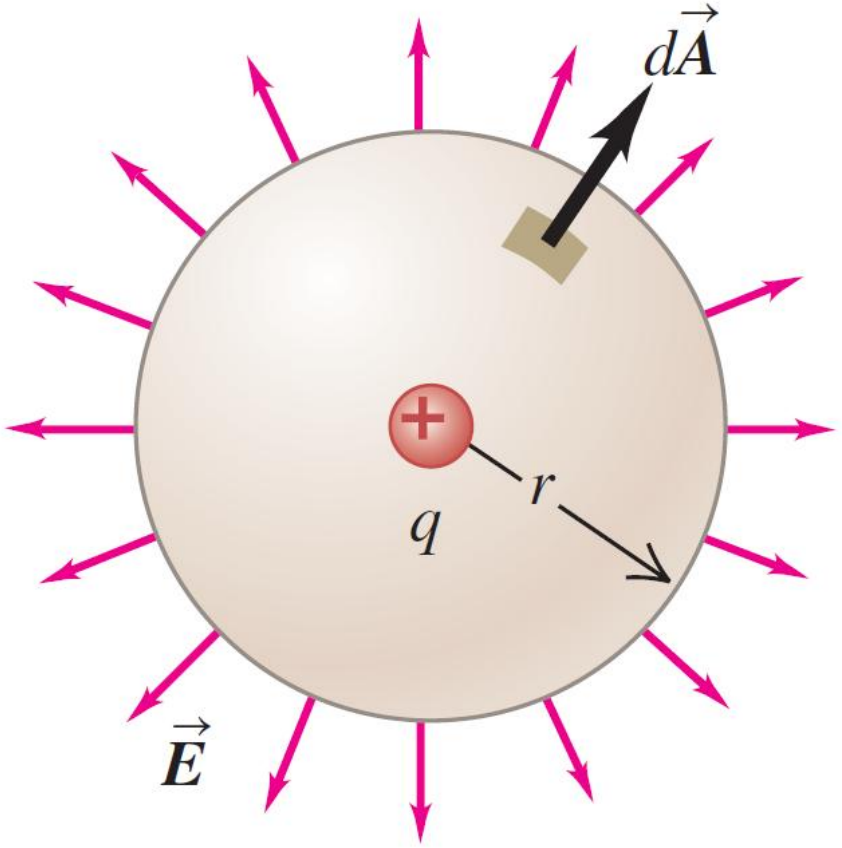
Evaluate

There is zero net flux of a uniform electric field through a closed surface that contains no electric charge.

Example 22.3 Electric flux through a sphere

A positive point charge $q = 3.0\ \mu C$ is surrounded by a sphere with radius $0.20\ m$ centered on the charge.

Find the electric flux through the sphere due to this charge.



Identify and Set Up

Here the surface is not flat and the electric field is not uniform, so we must use the general direction of electric flux.

Because the sphere is centered on the point charge, at any point on the spherical surface, $\vec{E}$ is directed out of the sphere perpendicular to the surface. The positive direction for both $\hat{n}$ and $E_\perp$ is outward, so $E_\perp = E$ and the flux through a surface element dA is $\vec{E} \cdot d\vec{A} = E dA$. This greatly simplifies the integral in Eq. (22.5).

Execute

At any point on the sphere the magnitude of $\vec{E}$ is

$$E = \frac{q}{4\pi\varepsilon_0 r^2}$$

$$= \left(9.0 \times 10^9\ N \cdot m^2/C^2\right)\frac{3.0 \times 10^{-6}\ C}{(0.20\ m)^2}$$

$$= 6.75 \times 10^5\ N/C$$

Because E is the same at every point, it can be taken outside the integral $\Phi_E = \int E dA$.

Execute

What remains is the integral $\int dA$, which is just the total area $A = 4\pi r^2$ of the spherical surface.

Thus the total flux out of the sphere is

$$\begin{aligned}\Phi_E &= EA \\ &= \left(6.75 \times 10^5\ N/C\right)(4\pi)(0.20\ m)^2 \\ &= 3.4 \times 10^5\ N \cdot m^2/C\end{aligned}$$

Evaluate

Notice that we divided by $r^2 = (0.20\,m)^2$ to find E, then multiplied by $r^2 = (0.20\,m)^2$ to find Φ_E; hence the radius r of the sphere cancels out of the result for Φ_E.

We would have obtained the same flux with a sphere of radius $2.0\,m$ or $200\,m$. We came to essentially the same conclusion from Fig. 22.4 in Section 22.1, where we considered rectangular closed surfaces of two different sizes enclosing a point charge.

Evaluate

There we found that the flux of $\vec{E}$ was independent of the size of the surface.

Indeed, the flux through any surface enclosing a single point charge is independent of the shape and size of the surface.

The electric field $\vec{E}$, or electric force per unit charge, is useful because it does not depend on the charge of the body on which the electric force is exerted.



- The electric force experienced by a test charge q_0 can vary from point to point, so the electric field can also be different at different points.

- For this reason, Eq. (21.4) can be used only to find the electric force on a point charge, not on large bodies where the electric field may vary with different locations on the body.

22.3 Gauss's Law

- **Gauss's law** states the total electric flux through any closed surface (a surface enclosing a definite volume) is proportional to the total (net) electric charge inside the surface. The magnitude E of the electric field at every point on the surface is given by

$$E = \frac{1}{4\pi\epsilon_0}\frac{q}{R^2}$$

22.3 Gauss's Law

- At each point on the surface, $\vec{E}$ is perpendicular to the surface, and its magnitude is same at every point, just as in Example 22.3.

- The total electric flux is just the product of the field magnitude E and the total area $A = 4\pi R^2$ of the sphere:

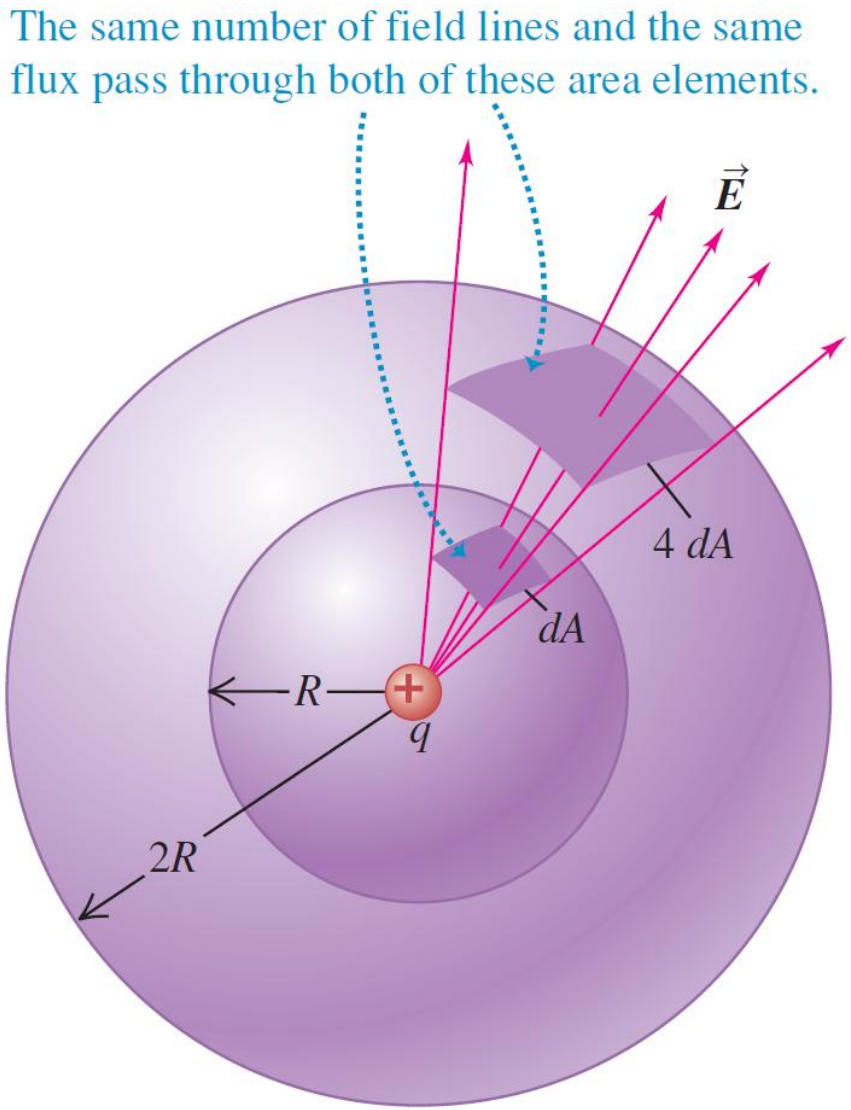
$$\Phi_E = EA = \frac{1}{4\pi\varepsilon_0}\frac{q}{R^2}\left(4\pi R^2\right) = \frac{q}{\varepsilon_0}$$

22.3 Gauss's Law

- The flux is independent of the radius R of the sphere. It depends only on the charge q enclosed by the sphere.
- We can also interpret this result in terms of field lines.
- The figure in the next slide shows two spheres with radii R and $2R$ centered on the point charge q.
- Every field line that passes through the smaller sphere also passes through the larger sphere, so the total flux through each sphere is the same.

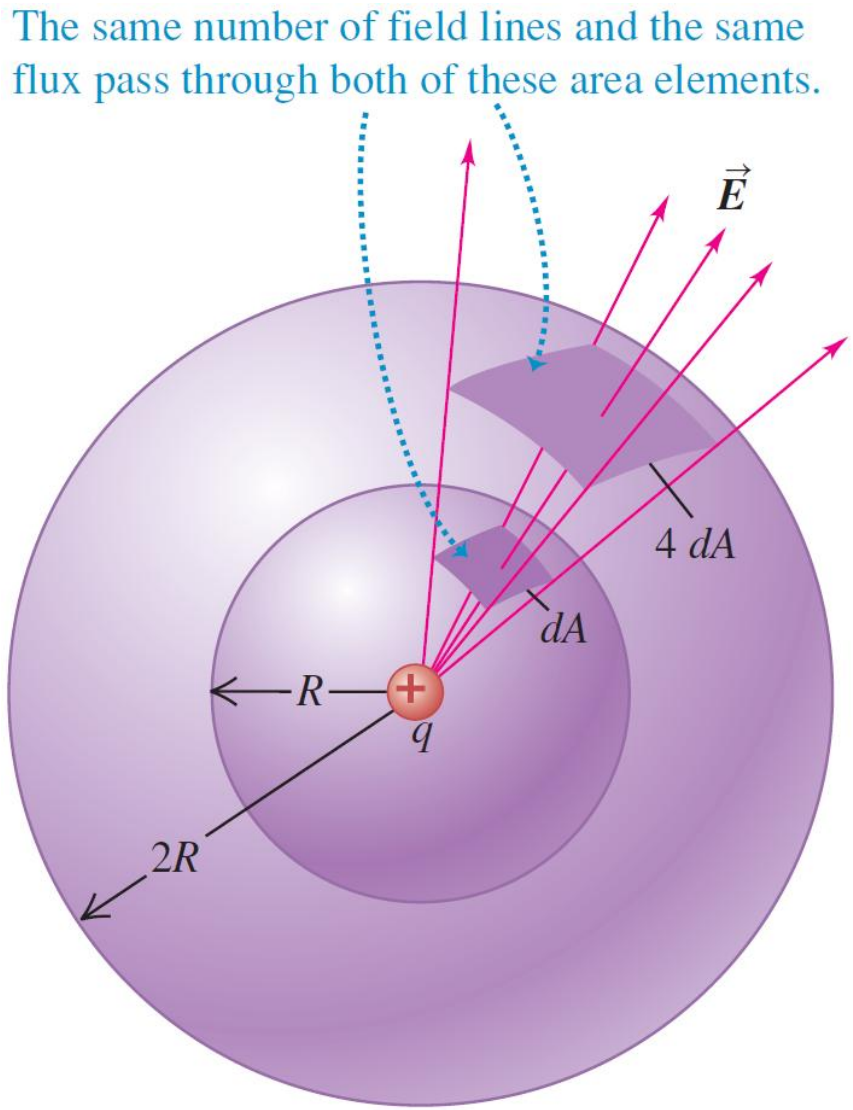
22.3 Gauss's Law

Projection of an element of area dA of a sphere of radius R onto a concentric sphere of radius $2R$. The projection multiplies each linear dimension by 2, so the area element on the larger sphere is $4\,dA$.



22.3 Gauss's Law

The projection multiplies each linear dimension by 2, so the area element on the larger sphere is $4\,dA$. The same number of field lines and the same flux pass through each area element.



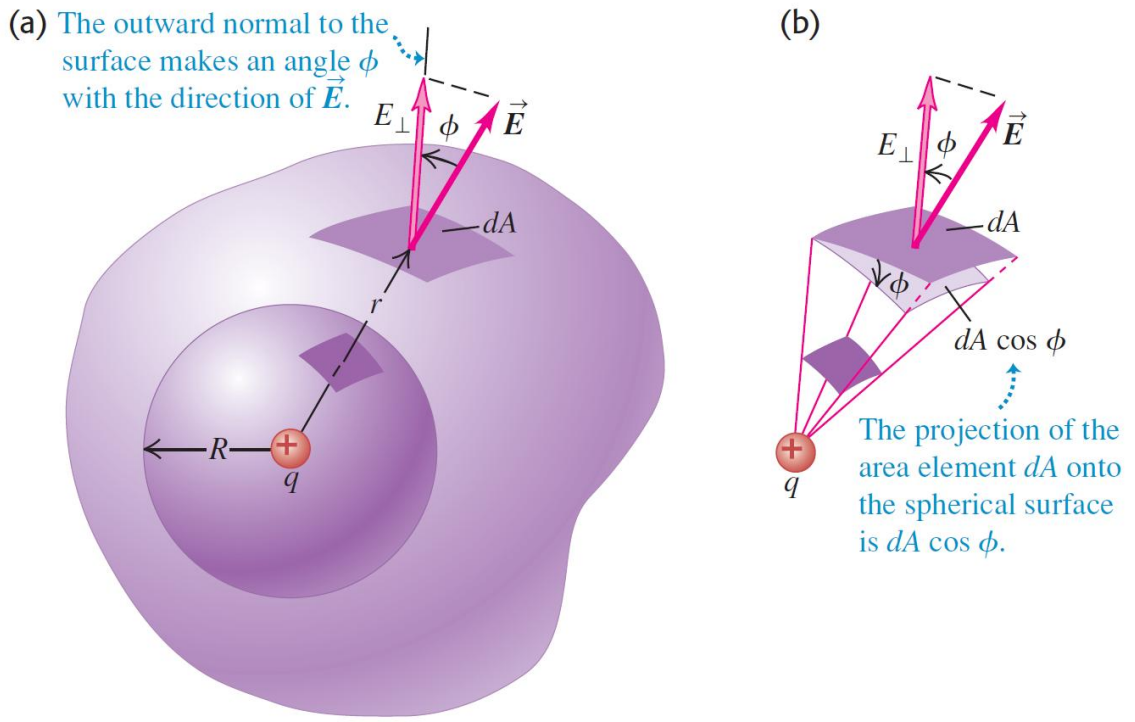
22.3 Gauss's Law

- For irregular surfaces, we can divide the entire irregular surface into elements dA , compute the electric flux $EdA\cos\phi$ for each, and sum the results by integrating.

- Each of the area elements projects onto a corresponding spherical surface element. Thus the total electric flux through the irregular surface must be the same as the total flux through a sphere, which Eq. (22.6) shows is equal to q/ε_0.

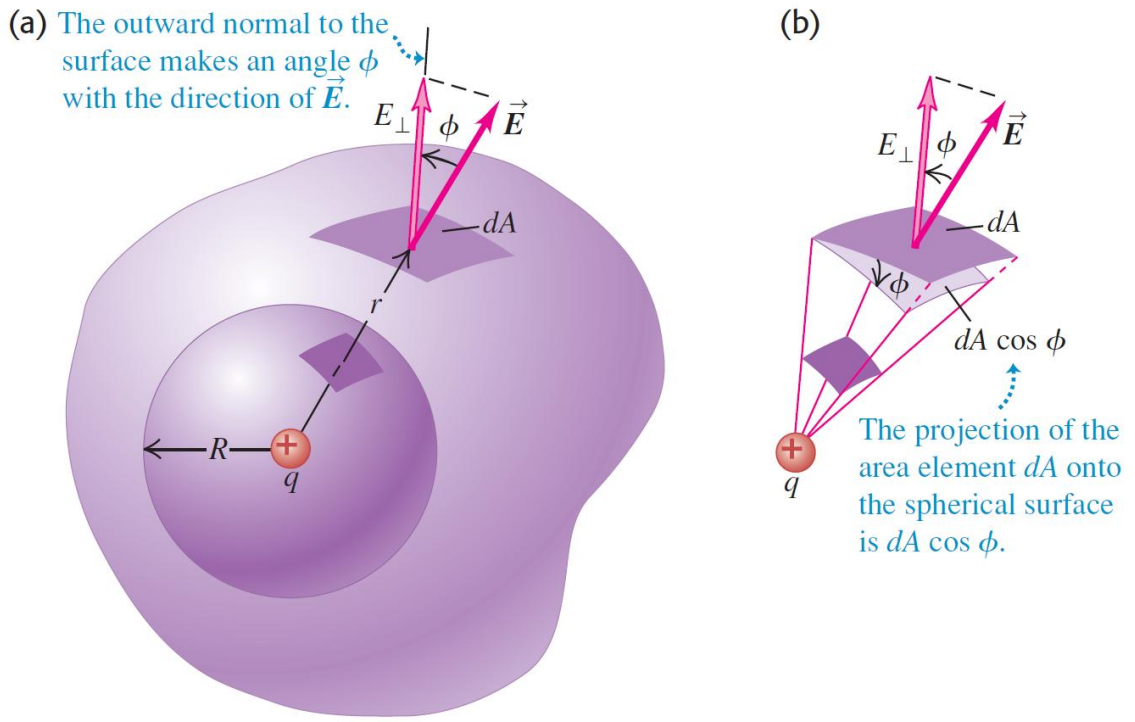
22.3 Gauss's Law

a) The outward normal to the surface makes an angle ϕ with the direction of $\vec{E}$.



22.3 Gauss's Law

b) The projection of the area element dA onto the spherical surface is $\mathrm{dA}\cos\phi$.



22.3 Gauss's Law

Thus, for the irregular surface:

$$\Phi_E = \oint \vec{\boldsymbol{E}} \cdot d\vec{\boldsymbol{A}} = \frac{q}{\varepsilon_0} \qquad (22.7)$$

- Equation (22.7) holds for a surface of any shape or size, provided only that it is a closed surface enclosing the charge q.
- The circle on the integral sign reminds us that the integral is always taken over a closed surface.

22.3 Gauss's Law

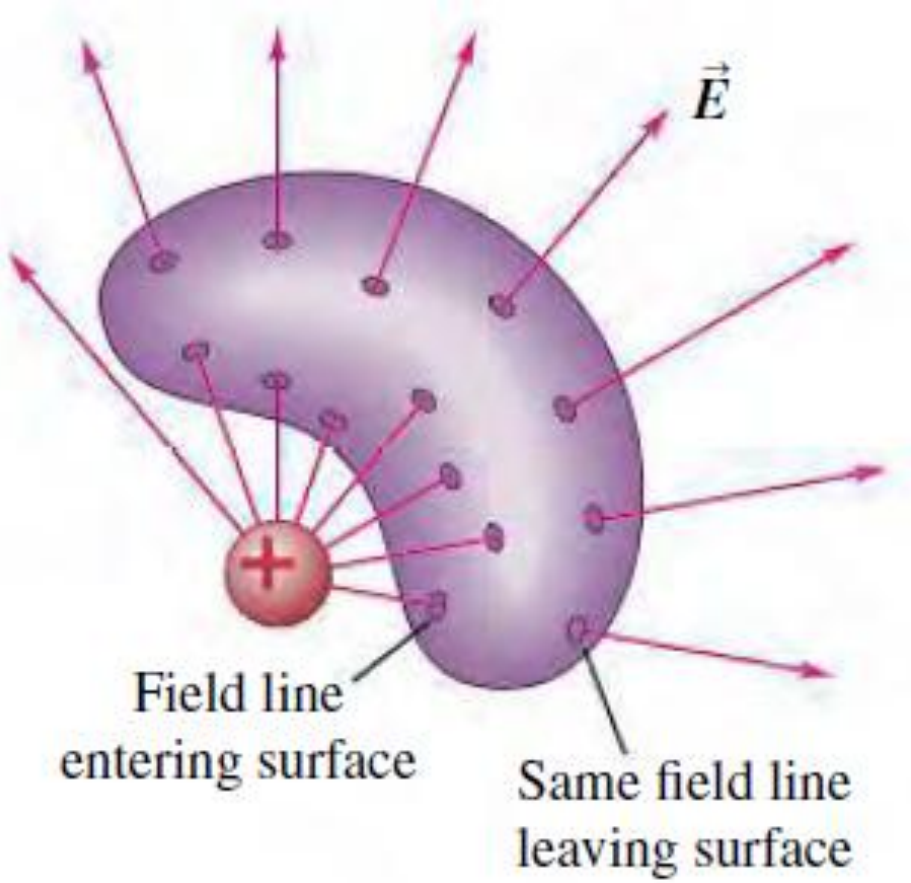
- The electric flux is then positive in area where the electric field points out of the surface and negative where it points inward.

- Also , $E_{\perp}$ is positive at points where $\vec{E}$ points out of the surface and negative at points where $\vec{E}$ points into the surface .

- Electric field lines can begin or end inside a region of space only when there is charge in that region which is shown in the figure below.

22.3 Gauss's Law

- A point charge outside a closed surface that encloses no charge.
- If an electric field line from the external charge enters the surface at one point, it must leave at another.



22.3 Gauss's Law

- The resultant electric field $\vec{E}$ at any point is the vector sum of the $\vec{E}$ fields of the individual charges.

- Let Q_{encl} be the total charge enclosed by the surface. Also let $\vec{E}$ be the total field at the position of the surface area element $\mathrm{d}\vec{A}$, and let E be its component perpendicular to the plane of that element (parallel to $\mathrm{d}\vec{A}$).

- We obtain the general statement of Gauss's law.

22.3 Gauss's Law

- We obtain the general statement of Gauss's law:

$$\Phi_E = \oint \vec{\boldsymbol{E}} \cdot d\vec{\boldsymbol{A}} = \frac{Q_{encl}}{\varepsilon_0} \qquad (Gauss's\ Law)$$

- The total electric flux through a closed surface is equal to the total (net) electric charge inside the surface, divided by ε_0.

$$\Phi_E = \oint E \cos\phi \, dA = \oint E_\perp \, dA = \oint \vec{\boldsymbol{E}} \cdot d\vec{\boldsymbol{A}} = \frac{Q_{encl}}{\varepsilon_0}$$



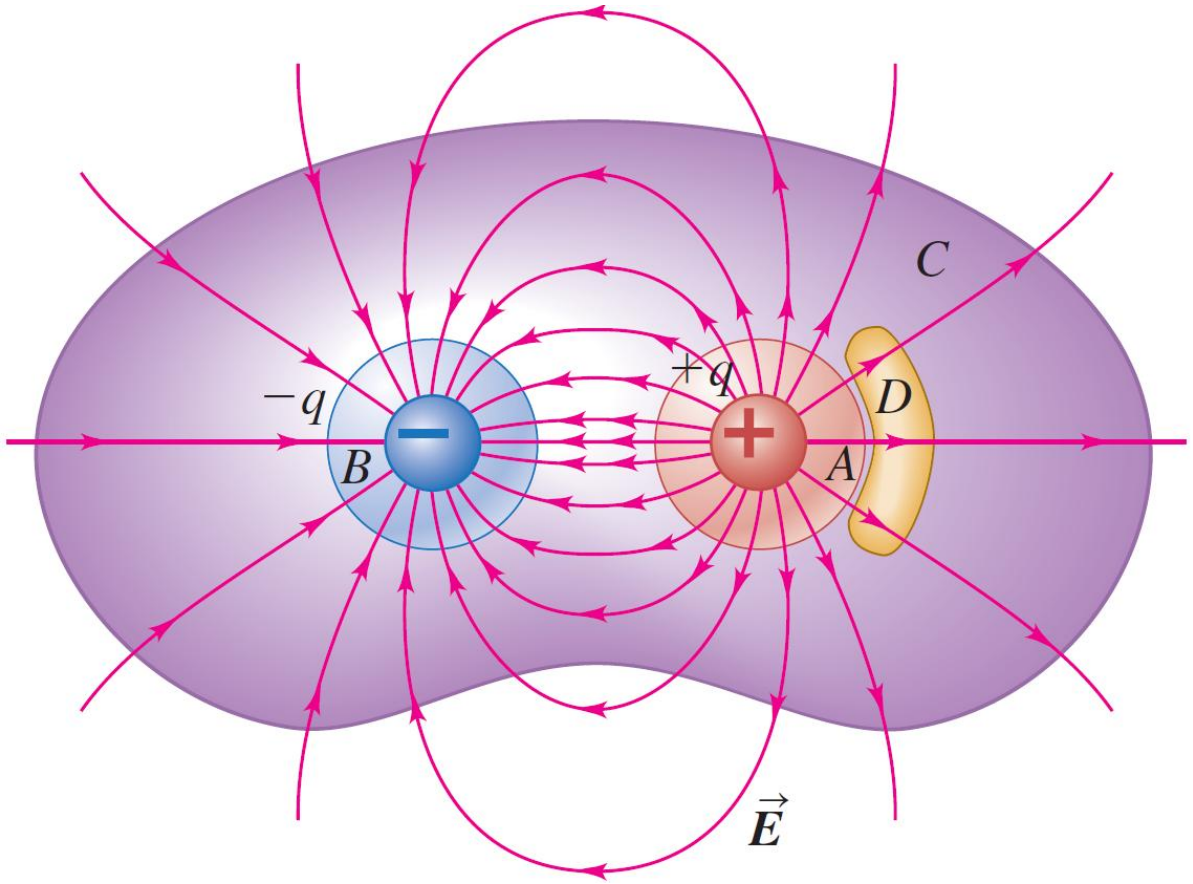
- Remember that the closed surface in Gauss's law is imaginary; there need not be any material object at the position of the surface.

- We often refer to a closed surface used in Gauss's law as a **Gaussian surface**.

Example 22.4 Electric flux and enclosed charge

The figure in the next slide shows that field produced by two point charges $+q$ and $-q$ of equal magnitude but opposite sign (an electric dipole).

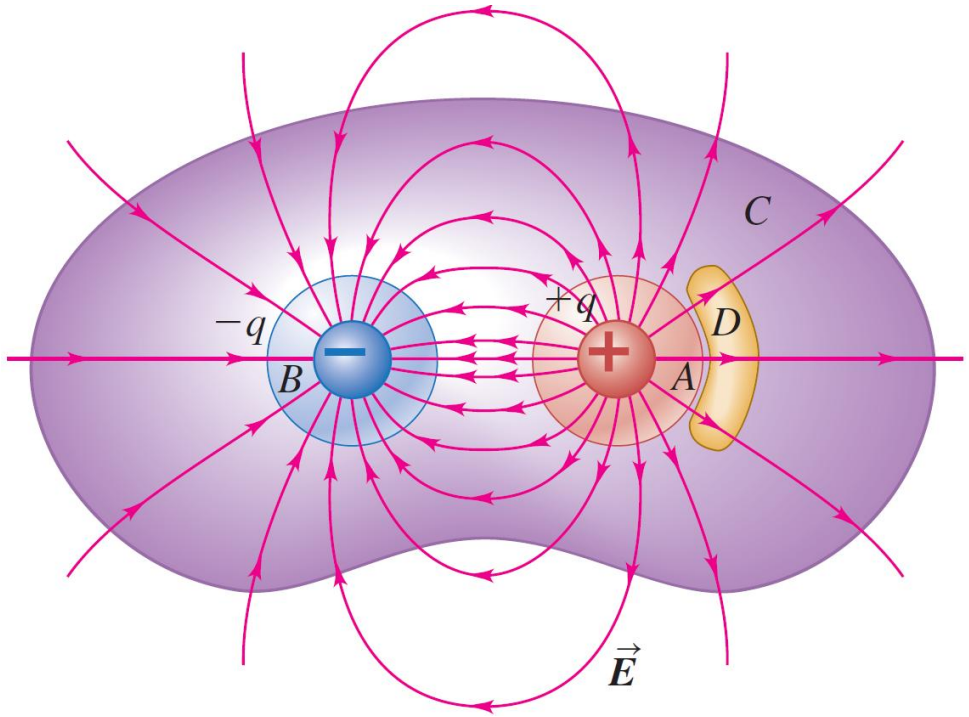
Find the electric flux through each of the closed surfaces A, B, C and D.



The net number of field lines leaving a closed surface is proportional to the total charge enclosed by that surface.

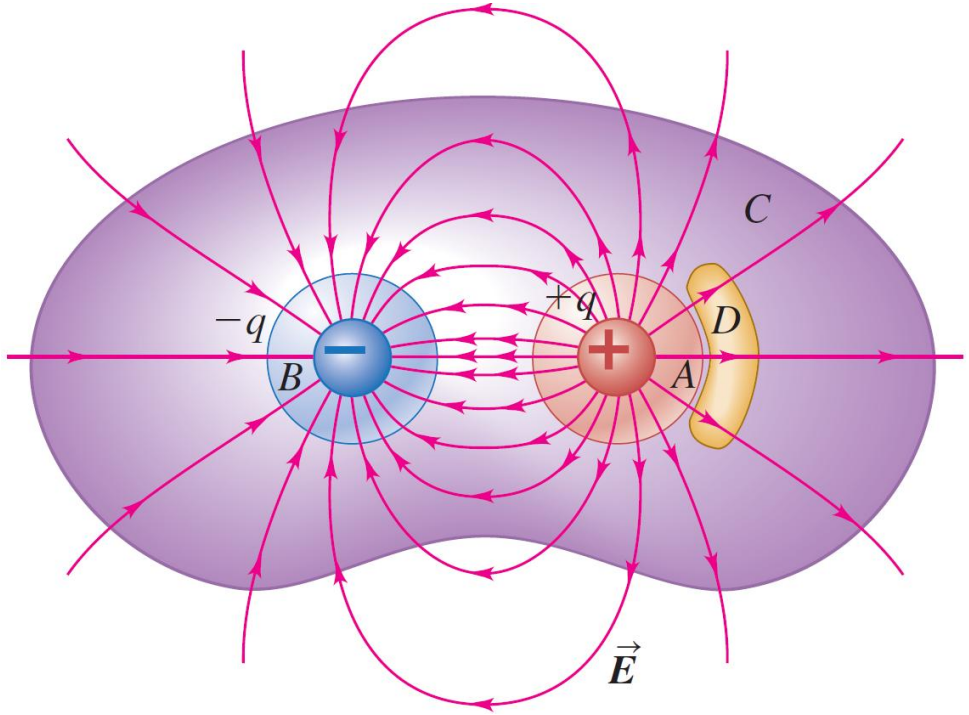
Solution

- Gauss's law says that the total electric flux through a closed surface is equal to the total enclosed charge divided by ε_0.



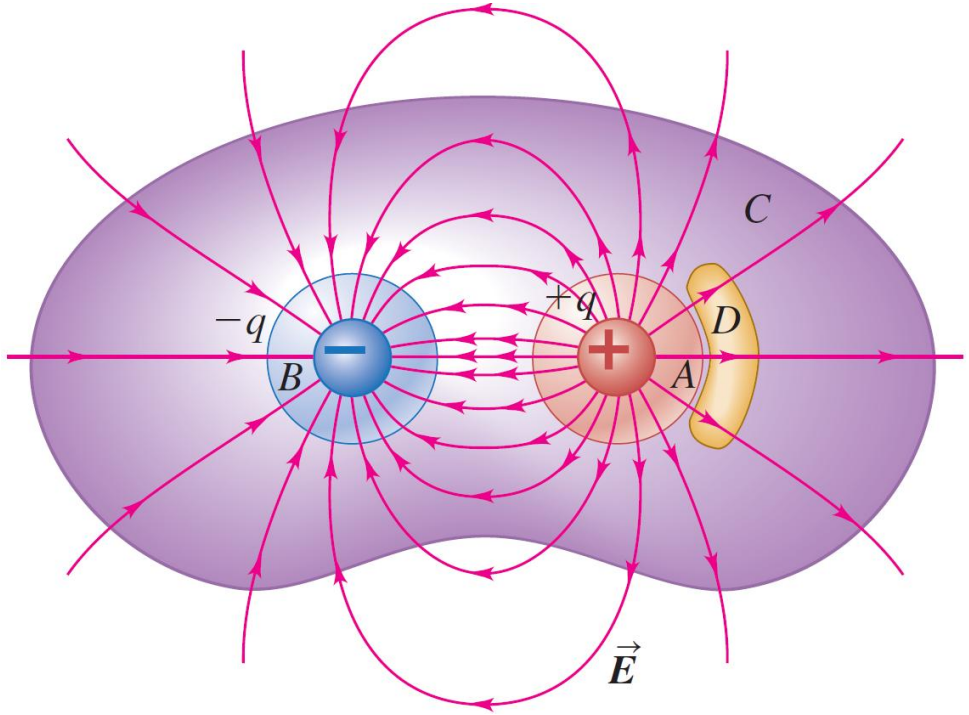
Solution

- By inspection of the figure, surface A (shown in red) encloses the positive charge, so $Q_{encl} = +q$.



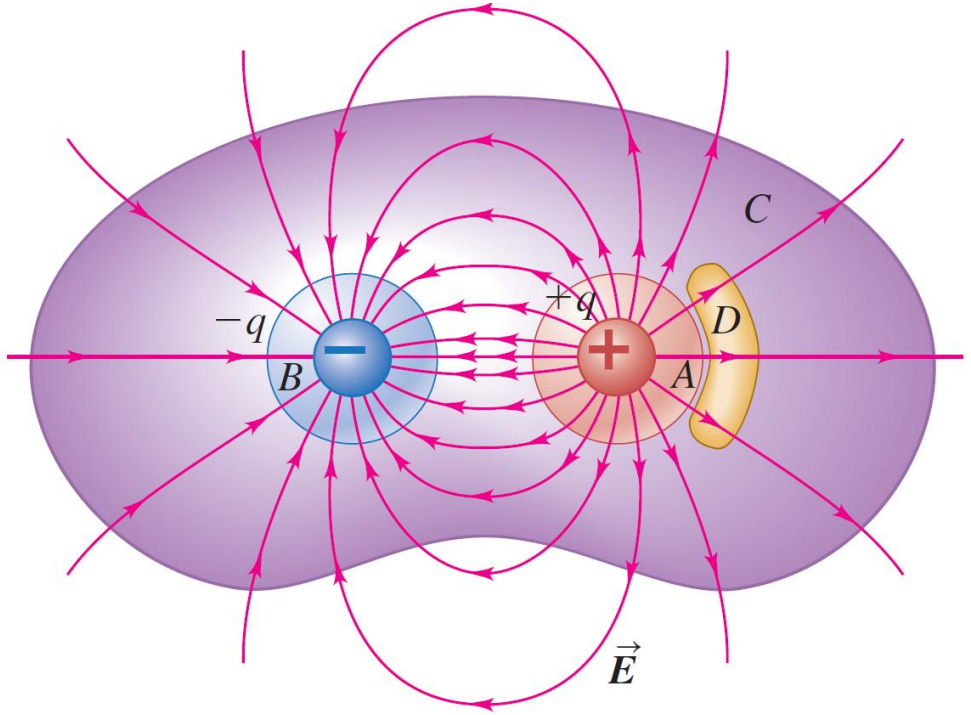
Solution

- Surface B (shown in blue) encloses the negative charge, so $Q_{encl} = -q$.



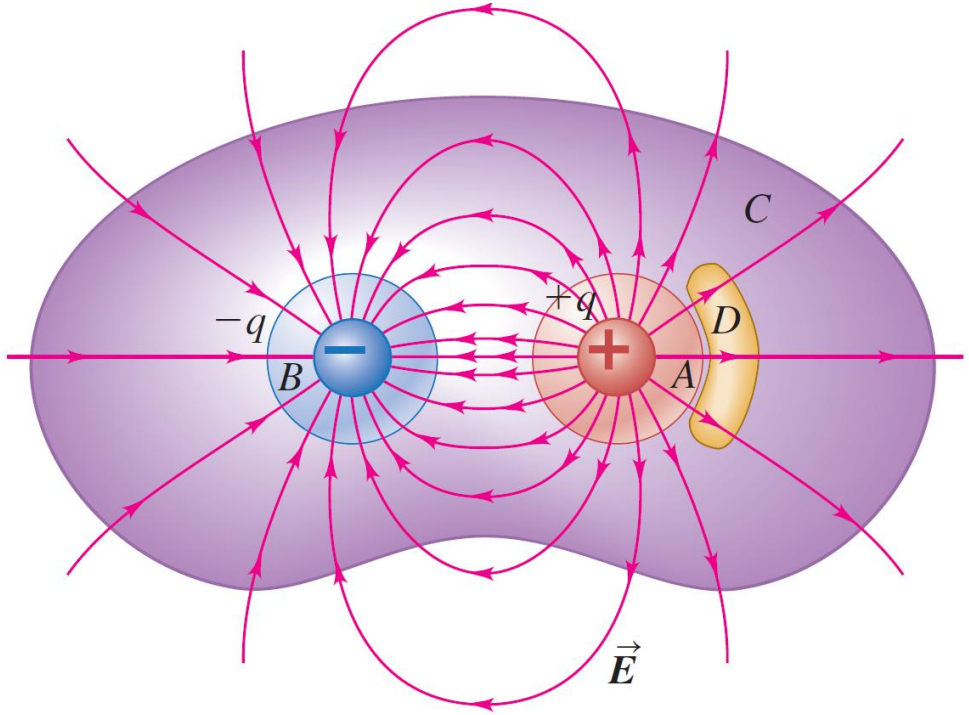
Solution

- Surface C (shown in purple), which encloses both charges, has $Q_{encl} = 0$.



Solution

- And surface D (shown in yellow), which has no charges enclosed within it, also has $Q_{encl} = 0$.



Solution

- Hence without having to do integration, we can conclude that the total fluxes for the various surfaces are $\Phi_E = +q/\varepsilon$ for surface A, $\Phi_E = -q/\varepsilon$ for surface B, and $\Phi_E = 0$ for surfaces C and D.

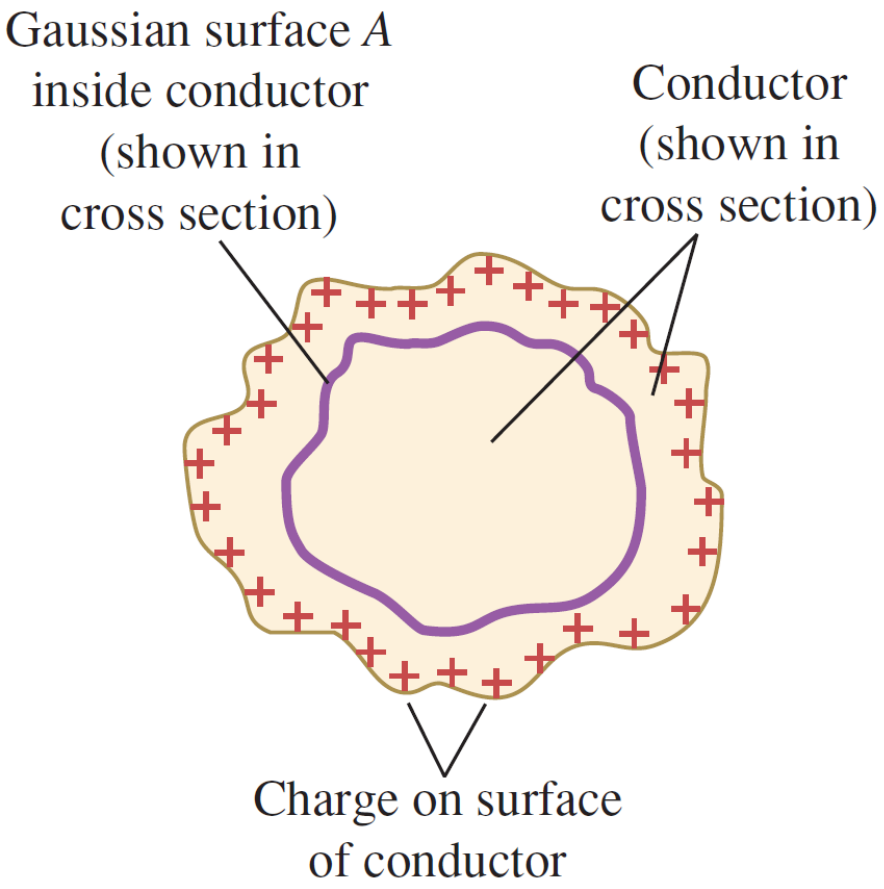
- Similar conclusions can be drawn by examining and comparing the electric field lines as well as the number of points of intersection between the surfaces and the field lines.

22.4 Applications of Gauss's Law

- Gauss's law is valid for any distribution of charges and for any closed surface.
- If we know the charge distribution, and if it has enough symmetry to let us evaluate the integral in Gauss's law, we can find the field.
- If we know the field, we can use Gauss's law to find the charge distribution, such as charges on conducting surfaces.

22.4 Applications of Gauss's Law

Under electrostatic conditions (charges not in motion), any excess charge on a solid conductor resides entirely on the conductor's surface.



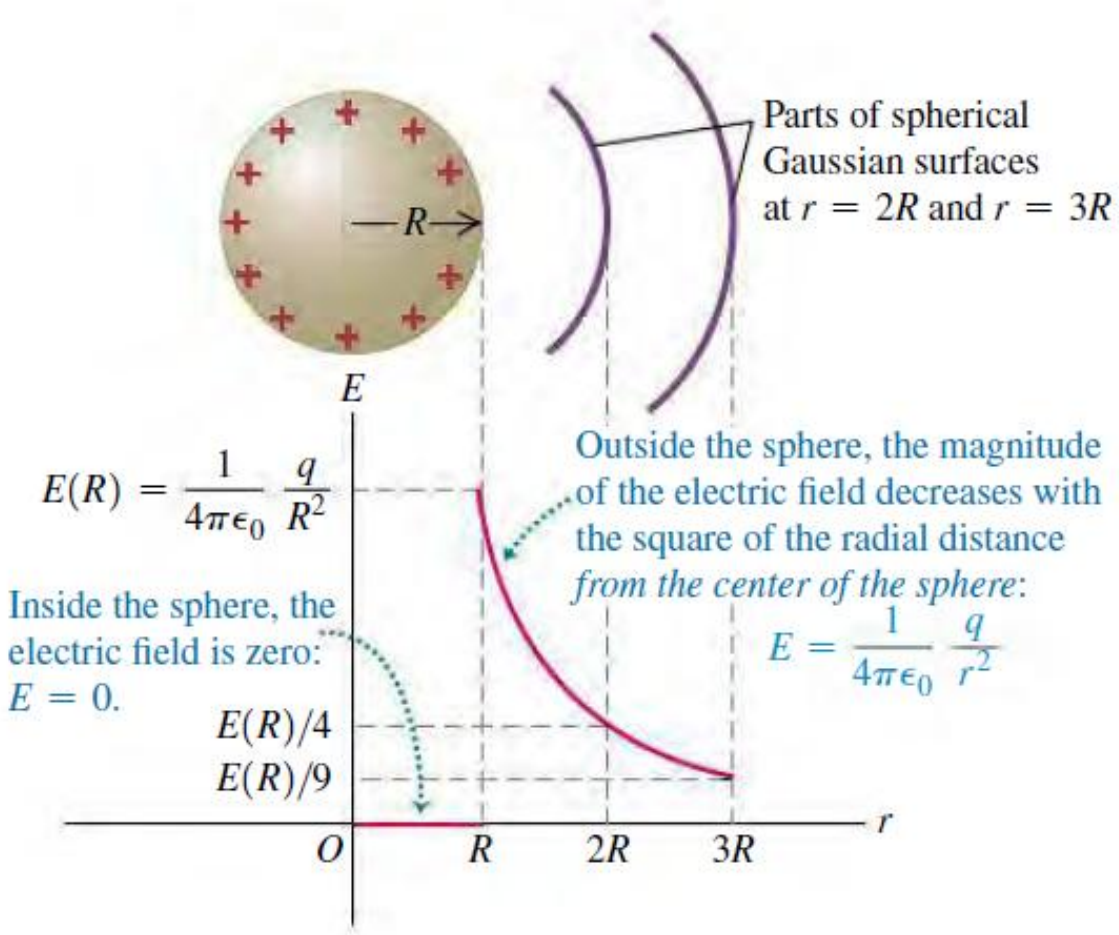
Example 22.5 Field of a charged conducting sphere

We place a total positive charge q on a solid conducting sphere with radius R (Figure in the next slide).

Find the electric field at any point inside or outside the sphere.

Under electrostatic conditions the electric field inside a solid conducting sphere is zero. Outside the sphere the electric field drops off as $1/r^2$, as though all of the excess charge on the sphere were concentrated at its center.

Example 22.5 (SOLN)



Identify

- As discussed earlier, all the charge must be on the surface of the sphere. The system has a spherical symmetry.

Set Up

- To take advantage of the symmetry, we take the Gaussian surface as an imaginary sphere of radius r centered on the conductor.

Execute

- A spherically symmetrical system after rotation is indistinguishable from the original unrotated system.

- To find $\vec{E}$ inside the conductor, we use a spherical Gaussian surface with radius $r < R$.

Evaluate

- In the limit as $R \to 0$, the sphere becomes a point charge; and the field is everywhere given by $E = q/4\pi\varepsilon_0 r^2$.

Example 22.6 Field of a line charge

Electric charge is distributed uniformly along an infinitely long, thin wire. The charge per unit length is λ (assumed positive).

Find the electric field.

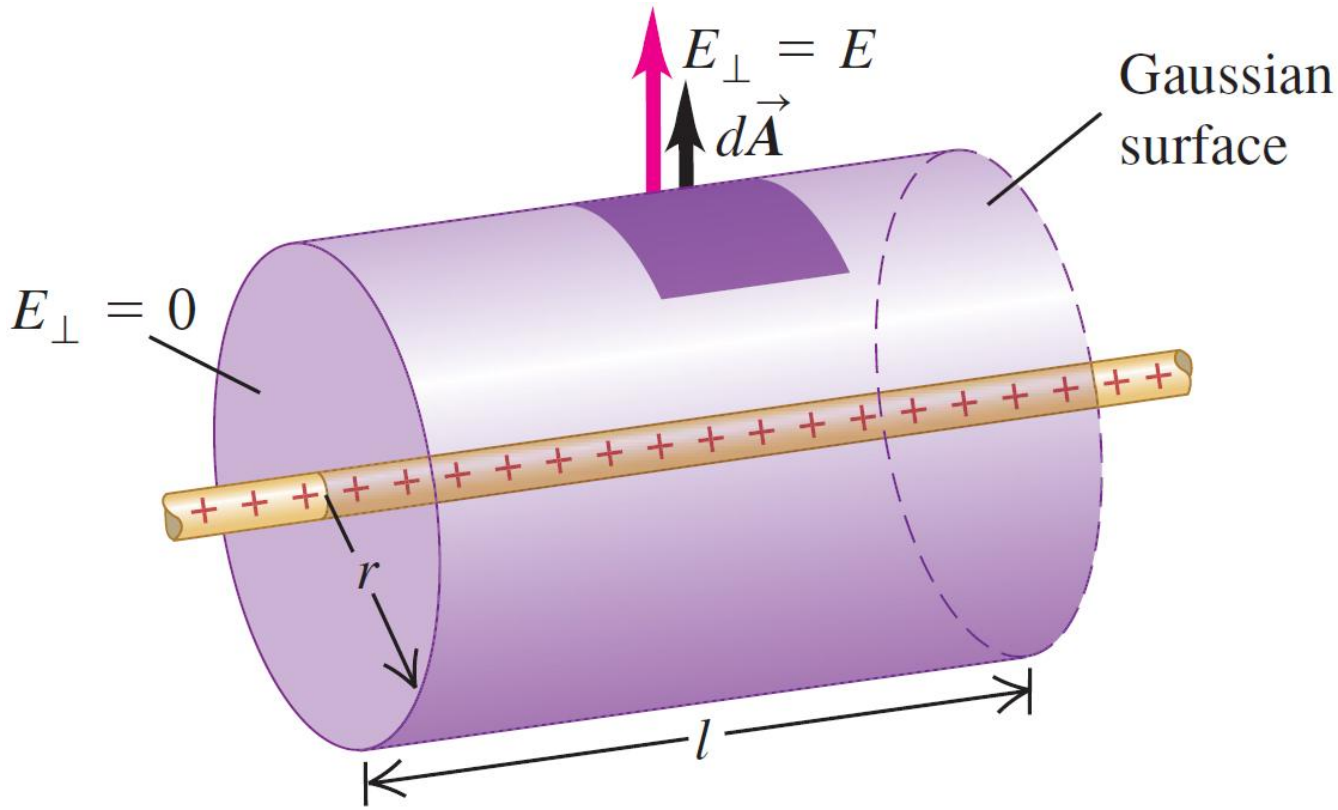
(This is an approximate representation of the field of a uniformly charged finite wire, provided that the distance from the field point to the wire is much less than the length of the wire.)

Identify

The system has cylindrical symmetry. The field must point away from the positive charges. To determine the direction of $\vec{E}$ more precisely, as well as how its magnitude can depend on position, we use symmetry as in Example 22.5.

Set Up

Use a Gaussian surface a cylinder with arbitrary radius r and arbitrary length l, with its ends perpendicular to the wire .



A coaxial cylindrical Gaussian surface is used to find the electric field outside an infinitely long charged wire.

Execute

From Gauss's law, Eq. (22.8),

$$\Phi_E = (E)(2\pi rl) = \frac{\lambda l}{\varepsilon_0}$$

and

$$E = \frac{1}{2\pi\varepsilon_0}\frac{\lambda}{r}$$

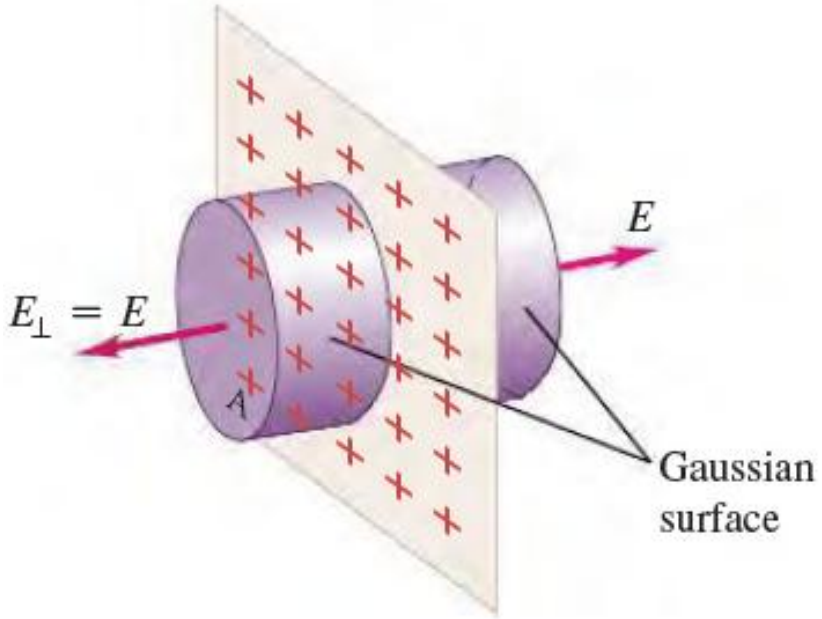
(field of an infinite line of charge)

Evaluate

Note that although the entire charge on the wire contributes to the field, only the part of the total charge that is within the Gaussian surface is considered when we apply Gauss's law.

Example 22.7 Field of an infinite plane sheet of charge

Find the electric field caused by a thin, flat, infinite sheet with a uniform positive surface charge density σ.



A cylindrical Gaussian surface is used to find the field of an infinite plane sheet of charge.

Identify

In this case, planar symmetry can be used to learn more about the direction and position dependence of $\vec{E}$.

Set Up

The charge distribution does not change in any direction parallel to the sheet. Use as the Gaussian surface a cylinder with its axis perpendicular to the sheet of charge, with ends of area A.

Execute

The charged sheet passes through the middle of the cylinder's length.

At each end of the cylinder, $\vec{E}$ is perpendicular to the charged sheet, it is parallel to the curved side walls of the cylinder, so $E_{\perp}$ at these walls is zero and there is no flux through these walls.

Thus the total flux integral in Gauss's law is $2EA$.

Execute

Taking $Q_{encl} = \sigma A$, Gauss's law, Eq. (22.8), gives

$$2EA = \frac{\sigma A}{\varepsilon_0} \quad and \quad E = \frac{\sigma}{2\varepsilon_0}$$

(field of an infinite sheet of charge)

The field is uniform and directed perpendicular to the plane of the sheet.

The field lines are therefore straight, parallel to each other, and perpendicular to the sheet.

Evaluate

The assumption we have made is that sheet is infinitely large.

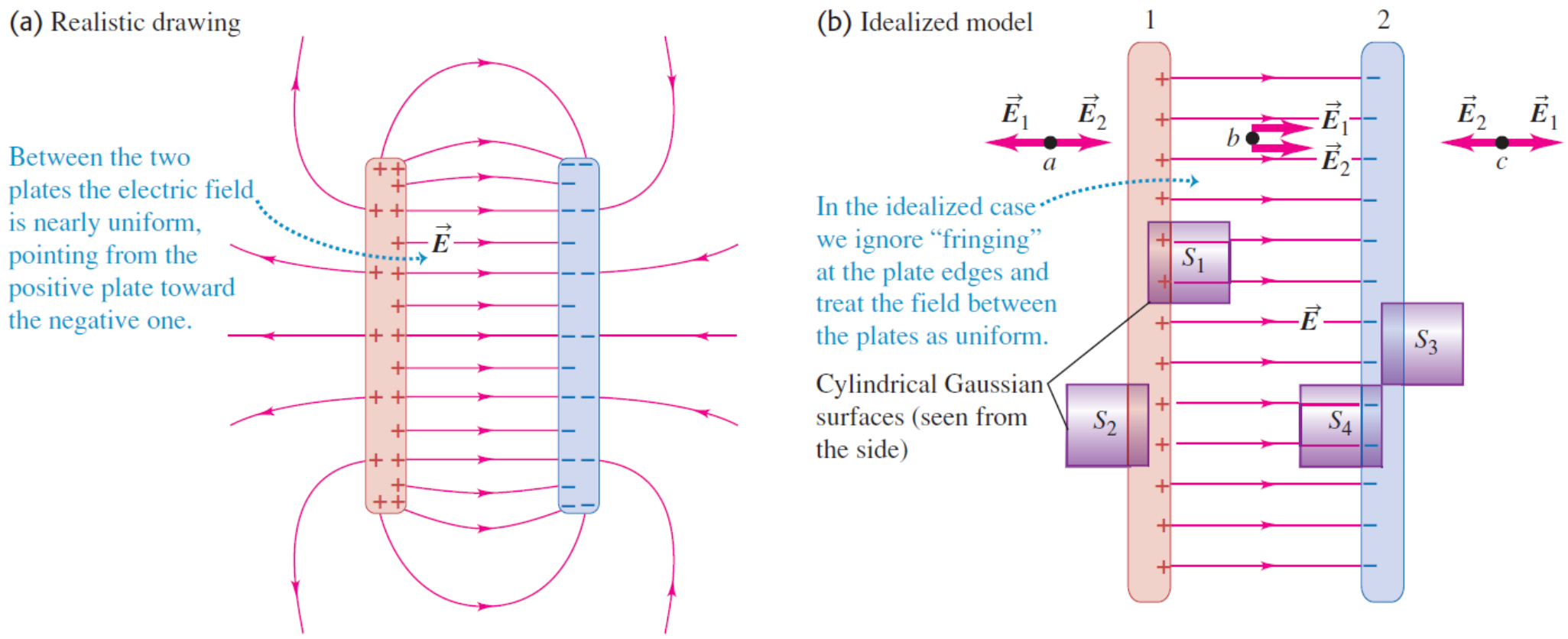
The result $E = \sigma/2\varepsilon_0$ is a good approximation for points that are close to the sheet (compared to the sheet's dimensions) and not too near its edges.

At such points, the field is very nearly uniform and perpendicular to the plane.

Example 22.8 Field between oppositely charged parallel conducting plates

Two large plane parallel conducting plates are given charges of equal magnitude and opposite sign; the charge per unit area is $+\sigma$ for one and $-\sigma$ for the other.

Find the electric field in the region between the plates. (Figure in the next slide)



Electric field between oppositely charged parallel plates (a) The field between the plates is quite uniform except near the edges. (b) idealized case in which "fringing" at the edges is ignored.

Identify

The field between and around the plates is approximately as shown in Fig. a.

Because opposite charges attract, most of the charge accumulates at the opposing facing of the plates. A small amount of charge resides on the outer surfaces of the plates, and there is some spreading or "fringing" of the field at the edges.

But if the plates are very large compared to the distance between them, the amount of charge on the outer surfaces is negligibly small, the "fringing" can be neglected except near the edges. (Fig. b)

Set Up

Use the shaded Gaussian surfaces S_1, S_2, S_3 and S_4. These surfaces are cylinders with ends of area A (refer to Fig. 22.21).

One end of each surface lies within one of the conducting plates.

Execute

For S_1, the left-hand end is within plate 1 (positive). There is no electric flux through this end.

The electric field between the plates is perpendicular to the right-hand end, hence $E_\perp = E$ and the flux is $+EA$ (out of surface).

No flux through the side walls (parallel to $\vec{\boldsymbol{E}}$).

Total flux integral in Gauss's law is EA .

Execute

The net charge enclosed by the cylinder is σA

$$EA = \frac{\sigma A}{\varepsilon_0}$$

and

$$E = \frac{\sigma}{\varepsilon_0}$$

(field between oppositely charged conducting plates)

Surface S_2, S_3, and S_4 can be analyzed similarly.

Evaluate

The above results were obtained using the principle of superposition of electric fields.

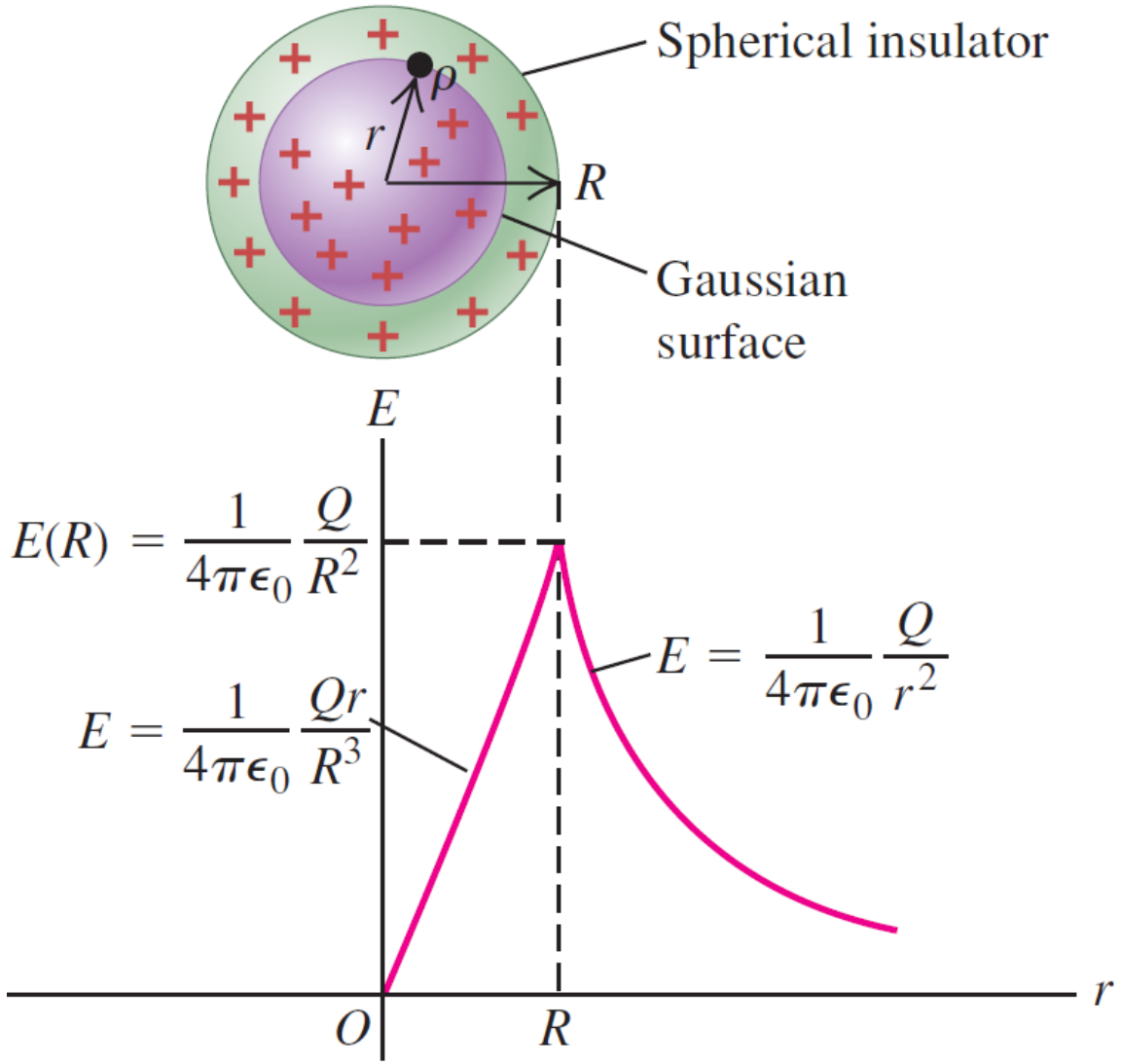
These results can be verified by using Gauss's law as well.

Example 22.9 Field of a uniformly charged sphere

Positive electric charge Q is distributed uniformly throughout the volume of an insulating sphere with radius R.

Find the magnitude of the electric field at a point P a distance r from the center of the sphere.

The following figure shows that the magnitude of the electric field of a uniformly charged insulating sphere. Compare this with the field for a conducting sphere.


Spherical insulator
ρ
r
R
Gaussian surface
E
$E(R) = \frac{1}{4\pi\epsilon_0}\frac{Q}{R^2}$
$E = \frac{1}{4\pi\epsilon_0}\frac{Qr}{R^3}$
$E = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}$
O
R
r

Identify

The system is spherically symmetric, hence the Gaussian surface used is a sphere with radius r concentric with charge distribution.

Execute

From symmetry, the total electric flux through the Gaussian surface is $\Phi_E = 4\pi r^2 E$.

The amount of charge enclosed within the Gaussian surface depends on the radius r.

Execute

- From symmetry, the total electric flux through the Gaussian surface is $\Phi_E = 4\pi r^2 E$. The amount of charge enclosed within the Gaussian surface depends on the radius r.

- Within the charged sphere, $r < R$, the field magnitude is proportional to the distance r of the field point from the center of the sphere.

- At the center, $r = 0, E = 0$.

Execute

- Outside the charged sphere, $r > R$, E varies as $1/r^2$.
- For any spherically symmetric charged body the electric field outside the body is the same as though the entire charge were concentrated at the center.
- Figure above shows a graph of E as a function of r for this problem.

Example 22.10 Field of a hollow charged sphere

A thin-walled hollow sphere of radius $0.250\,m$ has an unknown amount of charge distributed uniformly over its surface. At a distance of $0.300\,m$ from the center of the sphere, the electric field points directly toward the center of the sphere and has magnitude $1.80 \times 10^2\ N/C$.

How much charge is on the sphere?

Identify and Set Up

The charge distribution is spherically symmetric.

Use a spherical Gaussian surface that is concentric with the charge distribution and that passes through the point of interest at $r = 0.300\ m$.

Execute

- By Gauss's law, the flux is equal to the charge q on the sphere divided by ε_0. Solving for q, we find

$$q = -E\left(4\pi\varepsilon_0 r^2\right)$$

$$= -\left(1.80 \times 10^2 \frac{N}{C}\right)(4\,\pi) \times \left(8.854 \times 10^{-12} C^2/N \cdot m^2\right)(0.300\,m)^2$$

$$= -1.80 \times 10^{-9}\,C$$

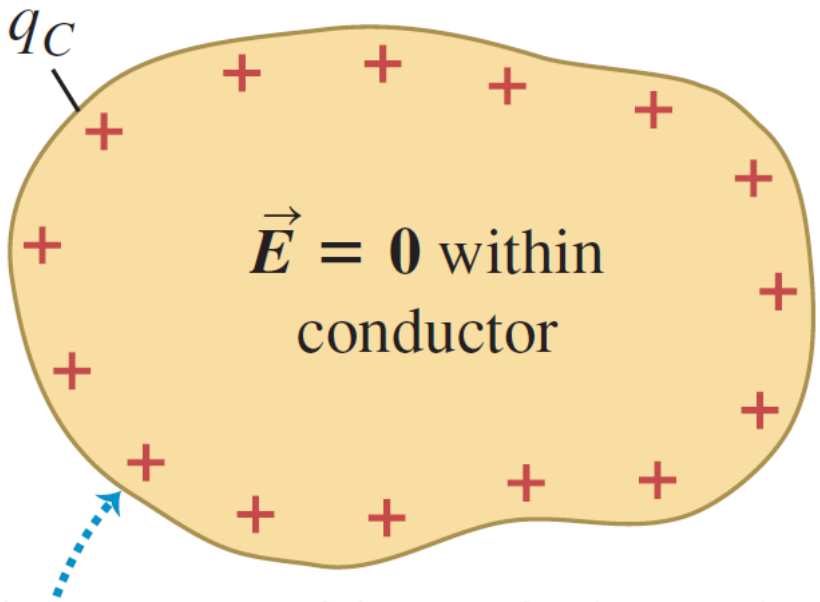
$$= -1.80nC$$

Evaluate

- We were able to calculate the flux integral since the electric field at all points on the Gaussian surface were known.
- This was possible because of the highly symmetric charge distribution.
- If the charge distribution is irregular or lacks symmetry, however, Gauss's law is not very useful for calculating the charge distribution from the field, or vice versa.

22.5 Charges on Conductors

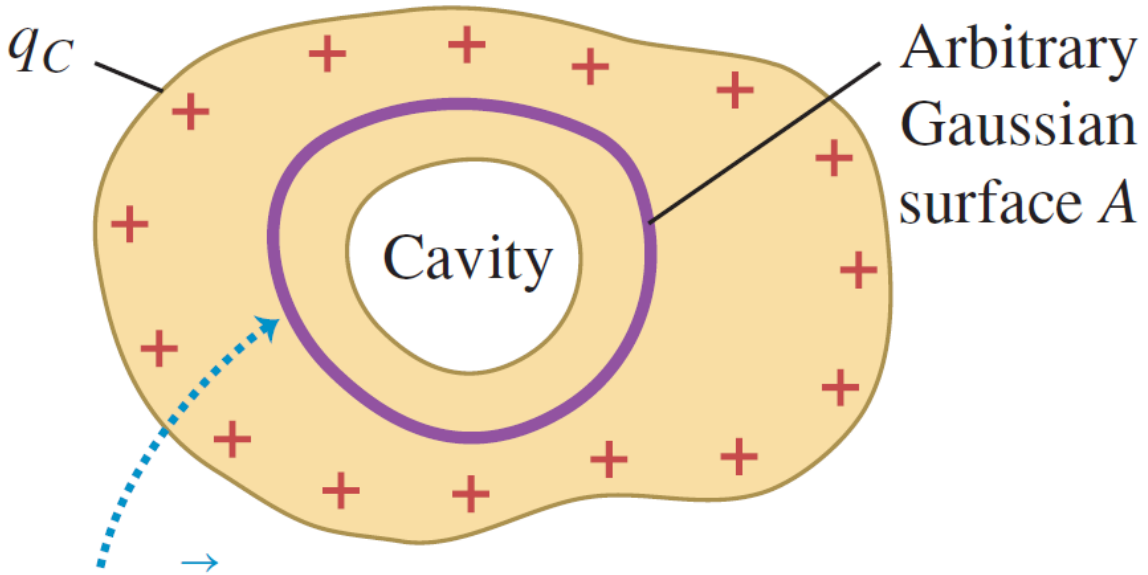
(a) Solid conductor with charge q_C



The charge q_C resides entirely on the surface of the conductor. The situation is electrostatic, so $\vec{E} = 0$ within the conductor.

22.5 Charges on Conductors

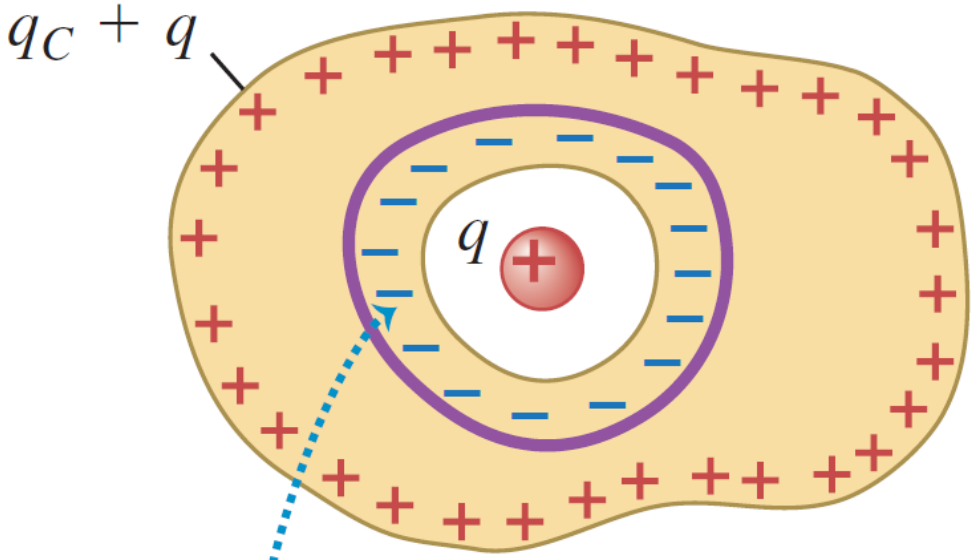
(b) The same conductor with an internal cavity



Because $\vec{E} = \mathbf{0}$ at all points within the conductor, the electric field at all points on the Gaussian surface must be zero.

22.5 Charges on Conductors

(c) An isolated charge q placed in the cavity



For $\vec{E}$ to be zero at all points on the Gaussian surface, the surface of the cavity must have a total charge $-q$.

22.5 Charges on Conductors

- If a small body with charge q is placed inside a cavity within a conductor (Fig. c) which is uncharged and insulated from the charge q, the total charge inside the Gaussian surface A is zero.

- A charge $-q$ is distributed on the inside surface of the cavity and thus attracts a charge $+q$ to form on the outer surface of the conductor.

22.5 Charges on Conductors

- By the same reasoning, the conductor originally had a charge q_c, then the total charge on the outer surface must be $q + q_c$, after the charge q is inserted into the cavity.

Example 22.11 A conductor with a cavity

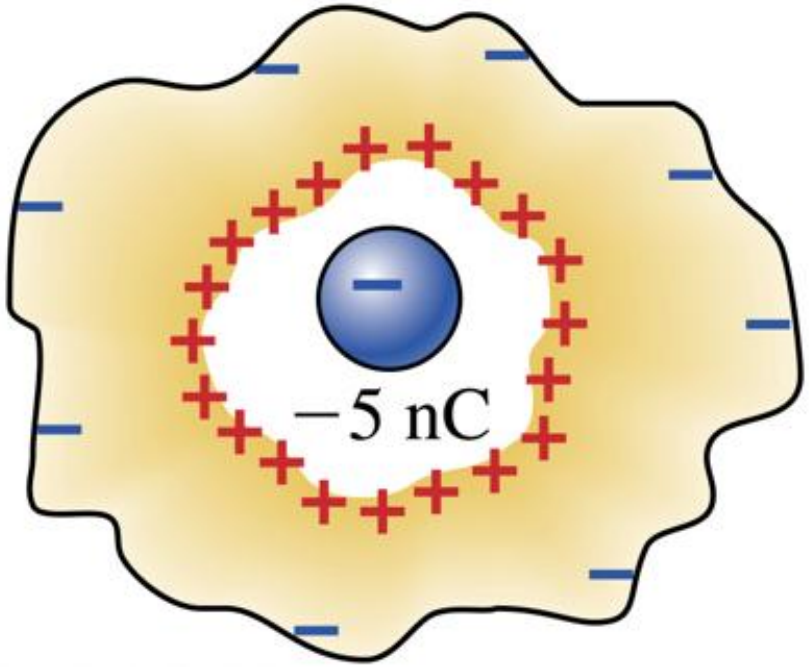
The conductor shown in cross section in the following figure carries a total charge of $+3\,nC$. The charge within the cavity insulated from the conductor, is $-5\,nC$.

How much charge is on each surface (inner and outer) of the conductor?

There is no excess charge in the bulk material of the conductor. Charge is located only on the inner and outer surfaces.

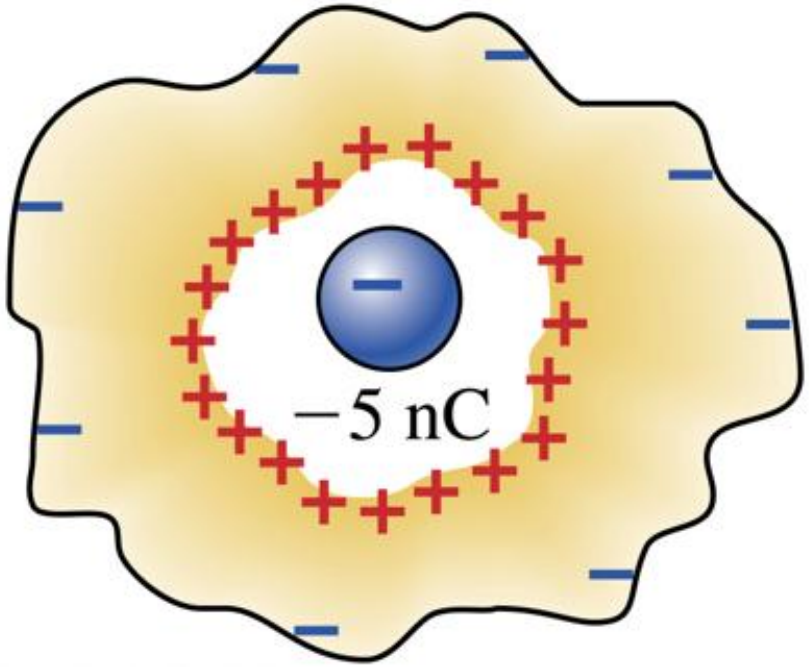
Solution

- If the charge in the cavity is $q = -5\,nC$, the charge on the inner cavity surface must be $-q = 5\,nC$.



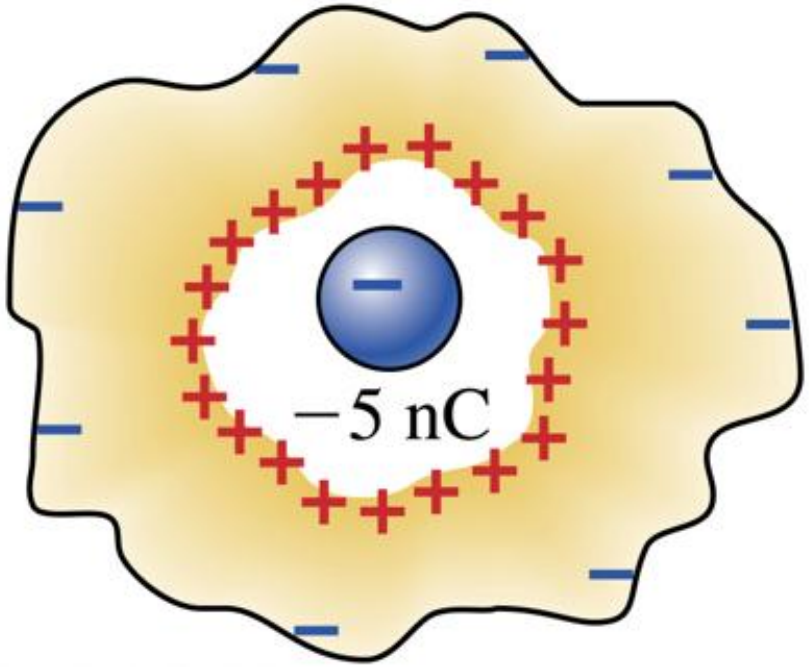
Solution

- The conductor carries a total charge of $+3\,nC$, none of which is in the interior of the material.



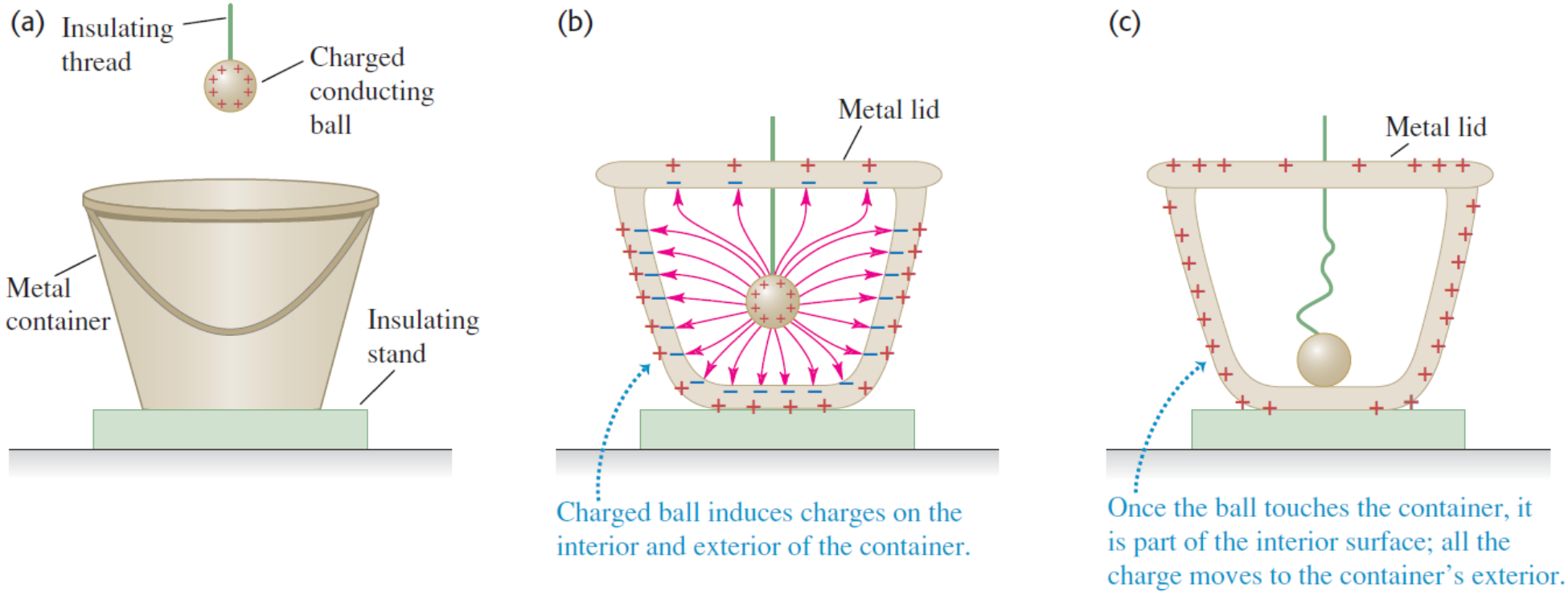
Solution

- If $+5\,nC$ is on the inner surface of the cavity, then there must be $(3\,nC)-(5\,nC) = -2\,nC$ on the outer surface of the conductor.



22.5 Charges on Conductors

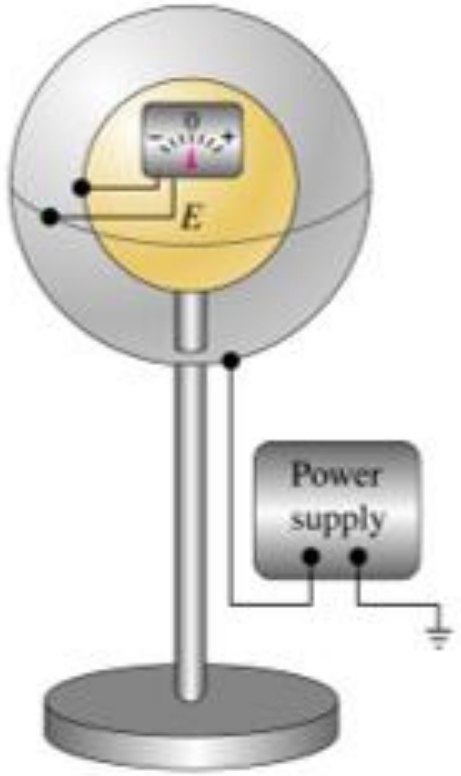
- The following two figures show Faraday's ice-pail experiment and a modern version respectively. Both experiments test Gauss's law.



22.5 Charges on Conductors

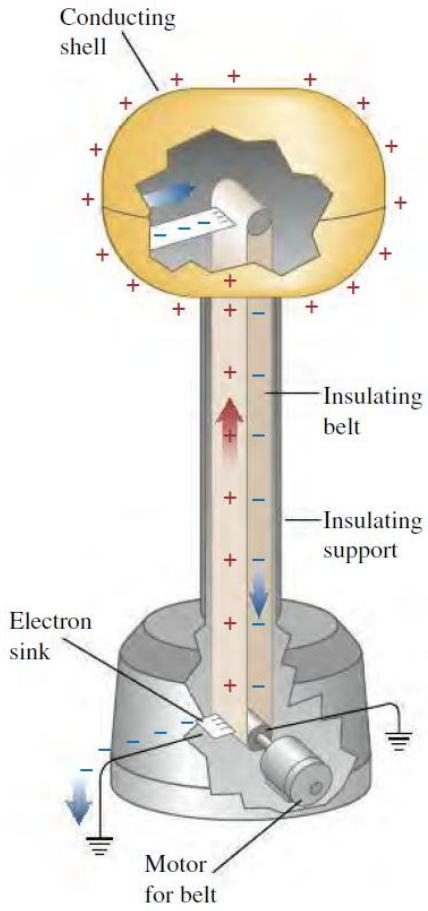
The outer spherical shell can be alternately charged and discharged by the power supply.

If there were any flow of charge between the inner and outer shells, it would be detected by the electrometer inside the inner shell.



22.5 Charges on Conductors

- The following figure shows a Van Graff generator used as an accelerator of charged particles and for physics demonstrations.
- Cutaway view of the essential parts of a Van Graff electrostatic generator.
- The electron sink at the bottom draws electrons from the belt, giving it a positive charge; at the top the belt attracts electrons away from the conducting shell, giving the shell a positive charge.



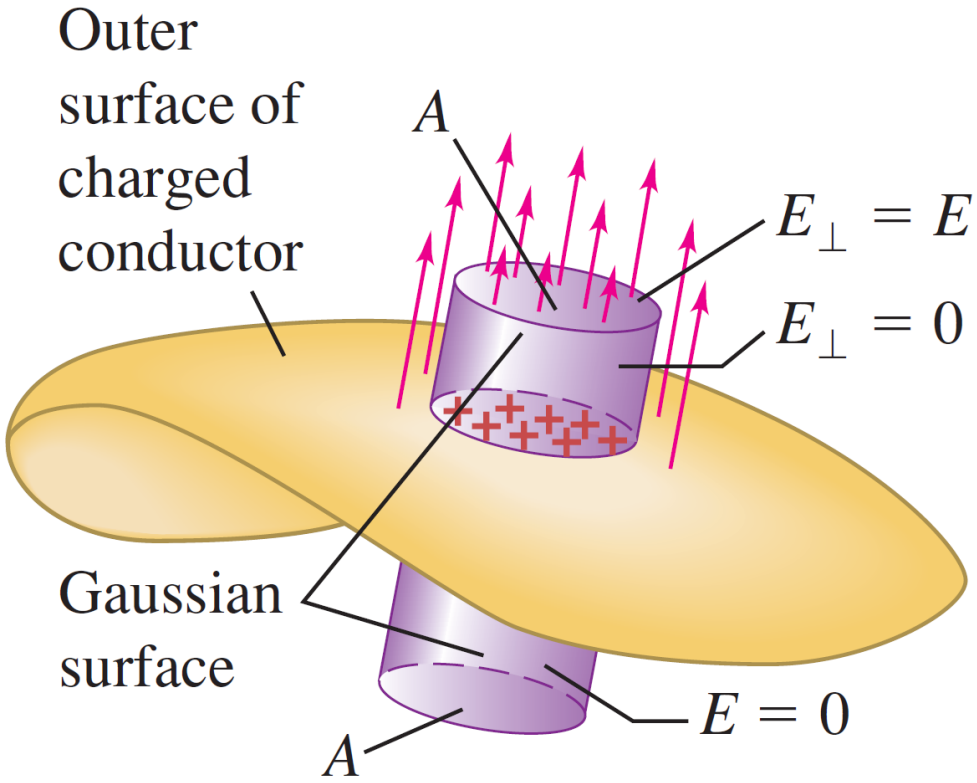
Field at the Surface of a Conductor

- There is a direct relation between the field $\vec{E}$ at a point just outside any conductor and the surface charge density σ at that point. The total flux through the surface is $E_{\perp}A$. The charge enclosed within the Gaussian surface is σA, so from Gauss's law,

$$E_{\perp}A = \frac{\sigma A}{\varepsilon_0} \quad and \quad E_{\perp} = \frac{\sigma}{\varepsilon_0} \quad (22.10)$$

(field at the surface of a conductor)

22.5 Charges on Conductors



The field just outside a charged conductor is perpendicular to the surface, and its perpendicular component $E_\perp$ is equal to σ/ε_0 .

Example 22.12 Field at the surface of a conducting sphere

Verify Eq. (22.10) for a conducting sphere with radius R and total charge q.

Solution

we showed that the electric field just outside the surface is:

$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{R^2}$$

Solution

The surface charge density is uniform and equal to q divided by the surface area of the sphere:

$$\sigma = \frac{q}{4\pi R^2}$$

Comparing these two expressions, we see that as Eq. (22.10) states, $E = \sigma/\varepsilon_0$.

Example 22.13 Electric field of the earth

The earth (a conductor) has a net electric charge. The resulting electric field near the surface can be measured with sensitive electronic instruments; its average value is about $150\ N/C$, directed toward the center of the earth

a) What is the corresponding surface charge density?

b) What is the total surface charge of the earth?

Identify and Set Up

Given the perpendicular electric field, we determine the surface charge density σ. The total surface charge on the earth is then the product ofσ and the earth's surface area.

Execute

a) We know from the direction of the field that σ is negative (corresponding to $\vec{E}$ being directed into the surface, so $E_\perp$ is negative).

$$\sigma = \varepsilon_0 E_\perp$$
$$= \left(8.85 \times 10^{-12}\ C^2/N \cdot m^2\right)(-150\ N/C)$$
$$= -1.33 \times (10^{-9} C)/m^2$$
$$= -1.33 nC/m^2$$

Execute

b) The earth's surface area is $4\pi {R_E}^2$, where $R_E = 6.38 \times 10^6\ m$, the radius of the earth. The total charge Q is the product $4\pi {R_E}^2 \sigma$, or

$$Q = 4\pi(6.38 \times 10^6\ m)^2\left(-1.33 \times 10^{-9}\ C/m^2\right)$$
$$= -6.8 \times 10^5\ C$$
$$= -680\ kC$$

Execute

Solving for Q, we find

$$Q = 4\pi\varepsilon_0 R^2 E_\perp$$

$$= \frac{1}{9.0 \times 10^9\ N \cdot m^2/C^2}(6.38 \times 10^6\ m)^2(-150\ N/C)$$

$$= -6.8 \times 10^5\ C$$

Evaluate

- One electron has a charge of $-1.60 \times 10^{-19}\ C$.
- Hence this much excess negative electric charge corresponds to there being about 7 moles of excess electrons.
- This is compensated by an equal deficiency of electrons in the earth's upper atmosphere, so the combination of the earth and its atmosphere is electrically neutral.

Concept Summary

Electric flux

- Electric flux is a measure of the "flow" of electric field through a surface.

- It is equal to the product of an area element and the perpendicular component of $\vec{E}$, integrated over a surface.

$$\Phi_E = \int E \cos\phi \, dA = \int E_\perp \, dA = \int \vec{E} \cdot d\vec{A} \qquad (22.5)$$

Gauss's Law

- Gauss's law states that the total electric flux through a closed surface, which can be written as the surface integral of the component of $\vec{E}$ normal to the surface, equals a constant times the total charge Q_{encl} enclosed by the surface.

- Gauss's law is logically equivalent to Coulomb's law, but its use greatly simplifies problems with a high degree of symmetry.

Gauss's Law

- When excess charge is placed on a solid conductor and is at rest, it resides entirely on the surface, and $\vec{\boldsymbol{E}} = 0$ everywhere in the material of the conductor.

$$\Phi_E = \oint E \cos\phi \, dA = \oint E_\perp \, dA = \oint \vec{\boldsymbol{E}} \cdot d\vec{\boldsymbol{A}} = \frac{Q_{encl}}{\varepsilon_0}$$

Electric field of various symmetric charge distributions

- The following lists electric fields caused by several symmetric charge distributions. Here, all the letter means the magnitudes of the quantities.

Charge Distribution	Point in Electric Field	Electric Field Magnitude
Single point charge q	Distance r from q	$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$
Charge q on surface of conducting sphere with radius R	Outside sphere, $r > R$	$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$
	Inside sphere, $r < R$	$E = 0$
Infinite wire, charge per unit length λ	Distance r from wire	$E = \frac{1}{2\pi\varepsilon_0}\frac{\lambda}{r}$

Charge Distribution	Point in Electric Field	Electric Field Magnitude
Infinite conducting cylinder with radius R, charge per unit length λ	Outside cylinder, $r > R$	$E = \frac{1}{2\pi\varepsilon_0}\frac{\lambda}{r}$
	Inside cylinder, $r < R$	$E = 0$
Solid insulating sphere with radius R, charge Q distributed uniformly throughout volume	Outside sphere, $r > R$	$E = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}$
	Inside sphere, $r < R$	$E = \frac{1}{4\pi\varepsilon_0}\frac{Qr}{R^3}$
Infinite sheet of charge with uniform charge per unit area σ	Any point	$E = \frac{\sigma}{2\varepsilon_0}$
Two oppositely charged conducting plates with surface charge densities $+\sigma$ and $-\sigma$	Any point between plates	$E = \frac{\sigma}{\varepsilon_0}$
Charged conductor	Just outside the conductor	$E = \frac{\sigma}{\varepsilon_0}$